

Inflation and Labor Markets: A Bottom-Up View

Sophia Chen* Deniz Igan[†] Do Lee[‡] Prachi Mishra[§]

Abstract

U.S. inflation surged in 2021-22 and has since declined, driven largely by a sharp drop in goods inflation, though services inflation remains elevated. This paper zooms into services inflation, using proprietary microdata on wages to examine its relationship with service sector wage growth at the Metropolitan Statistical Area (MSA) level. We estimate wage-price pass-through with a local projection instrumental variable model that exploits variation in labor market tightness across MSAs. Our findings reveal a positive, significant relationship between wages and price growth, with a lag, suggesting tight labor markets may have persistent effects and influence progress toward the inflation target.

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*Corresponding author. International Monetary Fund. 700 19th St NW, Washington DC, USA. Email: ychen2@imf.org

[†]International Monetary Fund and CEPR. Email: digan@imf.org

[‡]New York University. Email: dql204@nyu.edu

[§]Ashoka University. Email: prachi.mishra@ashoka.edu.in

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1 Introduction

U.S. inflation rose sharply in 2021-22, with year-on-year Consumer Price Index (CPI) inflation peaking at 9.0 percent in June 2022. Although it declined substantially over the subsequent year to 3.1 percent by June 2023, inflation has since proven persistent (Figure 1).

As dramatic as the movements in headline inflation have been, the divergence between price trends in goods and services has been equally striking. The initial surge in inflation in 2021 was driven by a spike in goods prices (Figure 2), resulting from supply-side disruptions in global supply chains and energy prices, along with a pandemic-induced shift in demand from services to goods. Services inflation began to rise later in the year as lockdowns eased and demand shifted back from goods to services. Although goods inflation peaked in early 2022 and its sharp decline has largely driven the moderation in overall inflation, services inflation did not peak until early 2023 and has remained sticky. Year-over-year services inflation stood at 4.4 percent at the end of 2024 and was still 3.3 percent at the end of 2025. Specifically, healthcare, transportation, and financial services continued to report elevated inflation. Therefore, understanding the drivers of services inflation is crucial to decoding the persistence of inflation.

In addition, the labor market has been central to the inflation debate. For instance, Bernanke and Blanchard (2024b) argue that, contrary to initial concerns that inflation would be driven by overly tight labor markets at the onset of the pandemic, the surge in inflation beginning in 2021 was largely due to price shocks—such as sharp increases in commodity prices and specific goods shortages—rather than wage pressures. In contrast, Ball et al. (2022) attribute a significant portion of the inflation rise to labor market tightness. Notably, labor market tightness has proven remarkably persistent, with the vacancy-to-unemployment ratio (V/U) still at 0.87 in December 2025 and averaging 1.01 over the preceding 12 months—substantially above the pre-pandemic historical average of 0.66. Although the labor market’s contribution to headline CPI inflation has diminished from its peak, it remains the largest estimated contributor to inflation above target, which could have negative implications for the outlook on headline inflation.¹

This paper develops a unified framework to explain these patterns in inflation and the labor market. It establishes an important role of labor market tightness in driving services inflation through wage growth channel. Specifically, we document a strong pass-through from local labor market tightness to wage growth in the service sector, and from service sector wage growth to services inflation. When the vacancy-to-unemployment ratio increases by one log point, services inflation tends to rise by 6.0 percentage points in the following months. These estimates are comparable to, though slightly larger than, recent estimates based on aggregate data. Additionally, we find some evidence of non-linearity, with higher estimates of wage-price pass-through when labor markets are tighter.

This paper distinguishes itself from the existing literature in three key respects. First, we focus specifically on explaining services inflation, which is particularly important given that persistently high services inflation continues to keep headline inflation well above the Federal Reserve’s 2 percent target. Second, we employ a novel identification strategy that exploits variation in labor market tightness across Metropolitan Statistical Areas (MSAs) using

¹This is based on the decomposition of Ball et al. (2022), extended to the most recent period.

a local projection instrumental variable (LP-IV) approach, allowing us to identify wage-price pass-through while addressing endogeneity concerns. Third, we leverage a proprietary micro dataset on service sector wages, combined with data from several additional sources, to provide one of the first U.S. microdata-based estimates of wage-price pass-through in the service sector.

Our conceptual framework is akin to a regional Phillips curve, capturing how local wage pressures driven by increased local demand translate into local inflation.² A key insight from the recent literature is that estimating the Phillips curve using cross-sectional data helps avoid biases from shifting long-run inflation expectations, which can distort the slope in time-series approaches. Moreover, this strategy allows us to isolate demand-driven wage pressures, addressing the confounding effects of simultaneous supply and demand shocks that typically influence the slope of the Phillips curve in opposite directions.

Our instrumental variable strategy broadly follows the intuition behind the tradable-demand instrument developed by Hazell et al. (2022), who estimate a regional Phillips curve using pre-pandemic data on state-level inflation and unemployment. To exploit variation in local demand for services, we instrument MSA-level service sector wage growth with a shift-share instrument based on national industry-level labor market tightness in the tradable sector (shift) and the employment share of those tradable industries within each MSA (share). This approach leverages the idea that supply or demand shocks in tradable sectors can affect local demand for non-tradable sectors differently, depending on an area’s exposure to the affected tradable sectors. For example, an oil price boom may boost demand for restaurant meals in Houston more than in Boston, due to Houston’s higher concentration of energy-related employment.³

Notably, a key innovation in our construction of the shift variable is that we use national changes in labor market tightness in the tradable sector only. Because prices in tradable sector are set in national or global markets, this measure is unlikely to be directly correlated with unobserved shocks to wage growth in local non-tradable (service) sectors. In other words, we assume that changes in labor market tightness in the tradable sector are driven by shocks that are specific to the tradable sector and uncorrelated with local supply shocks for non-tradables. Importantly, as positive (negative) shocks in the tradable sector increases (decreases) the demand for local non-tradables, firms in the non-tradable sector will experience an increase (decrease) in competition for workers, which ought to result in them raising (dropping) wages and prices.

A key input into our empirical methodology is a proprietary high-frequency wage dataset covering the service sector at the local market level. Local market data are essential for capturing cross-sectional variation in estimation of the regional Phillips curve, while high-frequency data are necessary to track the rapid changes in inflation seen in recent periods. However, few existing U.S. labor market datasets meet both criteria. For example, the Current Population Survey (CPS) and Current Employment Statistics (CES) lack sufficient geographic detail for cross-sectional analysis. The Quarterly Census of Employment and Wages (QCEW), on the other hand, offers geographic and industry-level tabulations, but

²See Hazell et al. (2022) for a review of the literature and theoretical foundations of the regional Phillips curve.

³We use non-tradable and service sectors interchangeable in the context of our IV estimation as nearly all services in the CPI calculation are non-tradable (Johnson, 2017).

micro data at the monthly level are not available.⁴

We address this gap by using a proprietary micro dataset from Homebase, a payroll service provider for small businesses in the services sector. The dataset includes detailed, high-frequency information on hours and wages for nearly 9 million workers across 1 million U.S. firms. Its granularity and frequency distinguish it from conventional datasets. Beyond filling a critical data gap, we demonstrate through external validity analyses that wage trends in the Homebase sample closely track official statistics. Further, we replicate our key findings using official data in place of the Homebase sample, but only under more restricted specifications—at lower frequency, with reduced granularity, and without directly measuring wage-price pass-through. Despite these limitations, the results are closely aligned. This alignment suggests that our findings, though drawn from a specific segment of the labor market, have broader relevance for understanding wage-inflation dynamics in the U.S. service sector as a whole. Importantly, the limitations of official data prevent them from fully capturing the mechanisms we document.

We use the vacancy-to-unemployment ratio as our primary measure of labor market tightness, following Ball et al. (2022) and Blanchard et al. (2022). Before the pandemic, the unemployment rate was the most common measure of labor market tightness, valued for its simplicity and availability. However, several researchers specializing in inflation dynamics, including Furman and Wilson (2021), Barnichon and Shapiro (2024), and Domash and Summers (2022), have argued that the vacancy-to-unemployment ratio better reflects labor market tightness in the post-pandemic economy, where labor force participation has fluctuated significantly. There is, however, now also a growing literature suggesting that the rise in vacancies may have been driven by inflation itself or changes in worker preferences towards more flexible work arrangements rather than labor market tightness (Afrouzi et al., 2024; Bagga et al., 2025; Guerreiro et al., 2024; and Pilossoph and Ryngaert, 2024). To mitigate the concern that this may be driving our results, we also use unemployment rate, employment growth, or wage growth as the shift variable to construct alternative instruments.

An first look at the data reveals an interesting pattern between services inflation and service sector wage growth at the aggregate level (Figure 3). Since mid-2021, these two series have shown a strong correlation with a lag, closely aligning with the timing of the rise in headline inflation.⁵

Our LP-IV estimates uncover an important role of wage growth in driving services inflation through the local labor market tightness channel. The first stage shows a strong pass-through from local labor market tightness to wage growth. Specifically, a one-log-point increase in the vacancy-to-unemployment ratio corresponds to a peak effect of a 12.5 percentage point increase in year-on-year service wage growth over a 10-month horizon. The

⁴The CPS, co-sponsored by the Census Bureau and the Bureau of Labor Statistics (BLS), surveys about 60,000 U.S. households and serves as the primary source for official unemployment statistics. The CES, sponsored by the BLS, surveys approximately 145,000 U.S. businesses and government agencies and provides official employment and wage statistics. The QCEW program by the BLS publishes a quarterly count of employment and wages reported by employers, covering more than 95 percent of U.S. jobs.

⁵The spikes in the wage growth series in early 2020 and early 2021 are consistent with aggregate data from the Current Employment Statistics (CES) (Stewart, 2022). They reflect changes in the composition of the workforce. In 2020, low-paid workers were laid off, pushing the average hourly wage artificially high. In 2021, as the economy reopened, employers filled many of these positions again.

second stage indicates that a one-percentage-point increase in service sector wage growth corresponds to a 0.48-percentage-point increase in year-on-year services inflation (excluding housing). Combining the estimates from the two stages, these results suggest that a one-log-point increase in the vacancy-to-unemployment ratio is associated with a 6.0-percentage-point increase in services inflation (excluding housing) over a 10-month horizon, operating through its effect on wage growth. Moreover, we find some evidence for the estimated effects to be nonlinear, with a higher wage-price pass-through, when labor markets are tighter, or the initial vacancy-to-unemployment ratio is higher. These findings are robust to alternative specifications and are broadly consistent with evidence from more aggregated and lower-frequency data.

These results indicate that local labor market tightness was a key driver of services inflation during the inflation surge period between Q3 2022 and Q1 2023, accounting for an average of 39.8 percent of services inflation (excluding housing) across MSAs during this period. Our findings also reinforce prior evidence from aggregate data, showing that the wage Phillips curve—or the relationship between wages and labor market tightness—is steeper under tighter labor market conditions.

These findings carry important implications for the ongoing policy debate. Even before the post-pandemic inflation surge, services inflation was traditionally the main driver of overall inflation in the United States. Services price growth tends to be more persistent than core goods price growth, largely due to the higher labor intensity of services, making them more sensitive to wage growth. Wage growth, in turn, is one of the most persistent input costs, reflecting the frequency of contract resets and other labor market settings and dynamics. The pandemic’s disproportionate impact on the services sector, particularly the difficulty in rehiring workers laid off during lockdowns, has contributed to overall labor market tightness. This highlights the critical role of wage trends in services and their impact on prices. While wage growth accelerated sharply after the pandemic’s acute phase and has recently moderated, wage pressures are likely to linger as purchasing power has yet to return to pre-pandemic levels.

Overall, our results suggest that a persistently tight labor market will exert persistent pressure on inflation. The effect of the pass-through from wages to service prices takes almost a full year to peak. This pressure may complicate the inflation outlook, especially in an environment prone to supply-side shocks due to geopolitical tensions, higher tariffs, and immigration restrictions. Taken together, the uncertainty around wage growth can feed into services inflation, and have medium to long run effects on core and headline consumer inflation.

The rest of the paper is organized as follows. Section 2 reviews the rapidly-expanding related literature. Section 3 provides an overview of the data and measurements. Section 4 lays out the empirical strategy. Section 5 presents the results. Section 6 concludes.

2 Related literature

Our paper is related to the growing literature that seeks to explain inflation dynamics in the post-pandemic period. This body of work has emphasized the importance of shocks to import prices and supply chains, with a natural focus on the goods sector due to its higher tradeability. For instance, Amiti et al. (2024) show that changes in import compe-

tition, along with a reduced ability to substitute between labor and intermediate inputs, contributed to the recent surge in inflation. Similarly, LaBelle and Santacreu (2022) highlight the significant impact of global supply chain disruptions on the U.S. Producer Price Index (PPI).

A separate strand of literature highlights the impact of tight domestic labor markets on price pressures. Ball et al. (2022) find that the high vacancy-to-unemployment ratios observed in 2021-2022 can explain a significant portion of the rise in monthly core inflation. They note that “the contribution of V/U to the rise in 12-month inflation is 2.0 percentage points, nearly a third of the total inflation increase. However, the rise in V/U explains more—nearly one-half—of the rise in core inflation and, the effect of V/U is rising over time.”

Dao et al. (2024) also find that tight labor markets continue to contribute significantly to inflation in the U.S., making it an exception among a large sample of advanced and emerging market economies, where inflationary pressure from relative price shocks has subsided, resulting in lower inflation. Bernanke and Blanchard (2024a) find an even larger quantitative role for V/U in explaining price pressures in the US compared to Dao et al. (2024). According to Dao et al. (2024), this difference stems from their inclusion of V/U in their equation for core inflation, while the approach of Bernanke and Blanchard (2024a) focuses solely on the relationship between V/U and wage inflation.

Our paper is closely related to the recent literature on Phillips curve non-linearity. Benigno and Eggertsson (2023) demonstrate that the Phillips curve slope steepens when the vacancy-to-unemployment ratio is high, attributing the recent inflation surge primarily to labor market tightness. Both Benigno and Eggertsson (2023) and Gitti (2024) provide empirical support for this non-linear relationship, finding that inflation accelerates rapidly once the vacancy-to-unemployment ratio exceeds unity. However, Beaudry et al. (2025) challenge the robustness of these findings. Analyzing both aggregate and city-level data, they show that alternative model specifications and data choices yield markedly different conclusions, suggesting that evidence for Phillips curve non-linearity remains fragile. We similarly find some evidence for non-linearity in wage-price pass-through, though it is not very robust.

Unlike the aforementioned papers which use the vacancy-to-unemployment ratio, Crump et al. (2022) rely on the unemployment rate as their measure of labor market tightness and complement it with multiple measures of labor compensation. They project underlying inflation to remain high due to tight labor markets and strong wage growth.⁶ Consistent with these papers, we find a strong role of the labor market in driving recent inflation. A key difference between our approach and these existing studies is that they examine time-series variation in aggregate inflation, while we explore cross-sectional variation across local markets. We provide a detailed comparison of the estimates in Section 5.

Our paper is also related to studies that estimate the Phillips curve using cross-sectional data (e.g., Beraja et al., 2019; McLeay and Tenreyro, 2019; Hooper et al., 2020). Hazell et al. (2022) show that using regional data helps overcome the issue of shifting long-run inflation expectations, which can confound the effect of labor market tightness. It also addresses the confounding effects of simultaneous supply and demand shocks that typically influence the

⁶Underlying inflation represents the inflation component that solely depends on the long-run trend and the sequence of current and future unemployment gaps.

slope of the Phillips curve in opposite directions. They show that the slope of the regional Phillips curve is steeper than that of an aggregate Phillips curve when labor market conditions are persistent. Most estimations in this literature focus on the pre-pandemic period. For example, Hazell et al. (2022) estimate the slope of the regional Phillips curve using data across U.S. states spanning 1978-2018. An exception is Barnichon and Shapiro (2024), who evaluate the performance of various tightness measures in predicting and explaining core inflation using MSA-level data at the quarterly frequency from 1982 to 2022. They find that the vacancy-to-unemployment ratio and vacancy filling cost proxies outperform other labor market tightness measures, such as the unemployment rate. Although both our paper and Barnichon and Shapiro (2024) estimate a cross-MSA Phillips curve, the focus and methodology differ substantially. Their analysis centers on the direct relationship between various measures of labor market tightness and inflation. In contrast, we focus specifically on the wage-price pass-through, using microdata from the services sector. Additionally, our analysis is conducted at the monthly level, providing a more granular view than their quarterly-level analysis.

Overall, our paper makes three contributions to the literature. First, we focus specifically on services inflation, a focus that is crucial given that elevated services inflation continues to keep headline inflation well above the Federal Reserve’s 2 percent target. The current policy debate centers on whether inflation has plateaued above this target, posing a potential challenge for monetary policy. As Bernanke and Blanchard (2024b) argue, the effects of overheated labor markets can be persistent, with labor market factors increasingly driving services inflation as the influence of goods prices wanes. This transmission is particularly strong in services because labor costs represent a larger share of total costs than in goods production, and staggered wage setting further amplifies the role of labor markets in sustaining inflation. Second, we employ a novel identification strategy that differs from existing approaches. We use a two-stage LP-IV framework in which we first identify labor market shocks using a Bartik instrument, and then estimate the pass-through from wages to prices. This approach effectively estimates wage-price pass-through by exploiting plausibly exogenous spatial variation in labor demand. Third, we leverage proprietary microdata on wages in the services sector, combined with data from several additional sources. By using high-frequency microdata within a cross-sectional Phillips curve framework, we can better identify the key drivers of persistent services inflation

3 Data and measurement

Our wage data come from a proprietary dataset provided by Homebase, a software company that offers scheduling, payroll reporting, and related services to businesses, primarily in the retail, hospitality, and other service sectors. The dataset is based on timecard records, offering detailed information on work hours and wages from nearly 9 million workers across 1 million firms U.S. firms. It includes granular daily employee-level data on hours worked, wages, job types, and links to the corresponding establishments and parent firms. Additionally, establishment-level details include location (via 5-digit zip codes) and industry classification (via 6-digit NAICS codes). Our sample period spans from January 2019 to

December 2022.⁷ We calculate average wages at the monthly frequency for each MSA.

With its high frequency and detailed coverage of work hours and wages at the micro level, the Homebase dataset stands out from conventional U.S. labor market sources, which lack sufficient geographic resolution or frequency for our purpose. However, the dataset has limitations, such as the exclusion of certain sectors and the lack of data on tips, benefits, and overtime payments. In addition, the sample of firms varies over time, as Homebase adoption among businesses changed during the sample period. Despite these limitations, we demonstrate through external validity analyses, discuss in detail in Section 5, that the wage and employment trends in the Homebase sample closely track official statistics from the CPS and CES. Additionally, changes in employment and earnings align well with CES data at the month-state level (Dvorkin and Isaacson, 2022; Chen and Lee, 2024).

For price data, we use CPI for 21 MSAs, sourced from the Bureau of Labor Statistics (BLS), which offers monthly or bi-monthly data at the MSA level. For MSAs with bi-monthly CPI data, we interpolate linearly to obtain a monthly frequency. We construct year-on-year inflation series at the MSA-month level. We focus on services inflation because nearly all CPI services are non-tradable (Johnson, 2017).⁸ This focus is important because prices set at the national level—typical for tradable goods—result in a flatter regional Phillips curve. We exclude housing inflation from our analysis because its drivers likely differ from those affecting non-housing services.⁹ Furthermore, the measurement of housing services inflation has unique challenges. Although much of housing expenditure comes from owner-occupied units, inflation is measured using rents for all shelter, since accurately deriving the user cost of housing from home prices is theoretically difficult. The literature has found that housing inflation exhibits a substantially different slope in the Phillips curve compared to non-housing services (Hazell et al., 2022; Stock and Watson, 2020), and that remote work has been a significant driver of post-pandemic housing prices (Howard et al., 2023).

4 Empirical approach

We analyze the data using an LP-IV approach, as used in Jordà et al. (2015) and Jordà et al. (2020). Local projections are a flexible and convenient method for estimating impulse responses (Jordà, 2005), which require minimal assumptions about the functional form of the responses. The LP-IV approach estimates these impulse responses to shocks using two-stage least squares. Specifically, we estimate:

$$y_{i,t}^h - y_{i,t}^0 = \alpha_t^h + \eta_i^h + \beta^h w_{i,t} + \sum_{k=1}^K \gamma_k^h w_{i,t-k} + \sum_{k=0}^K \delta_k^h y_{i,t-k} + \gamma^h X_{i,s,t} + u_{i,t}^h \quad (1)$$

⁷The sample on prices and wages starts a year earlier to allow for the calculation of annual changes.

⁸According to Johnson (2017), the only three CPI item codes for services classified as tradable are RA04 (Video cassettes, discs, and other media, including rentals), TF09 (unsampled motor vehicle fees), and TG01 (airline fare). Hazell et al. (2022) construct a non-tradable inflation series using BLS microdata, defining non-tradables similarly to the BLS service aggregation but with two exceptions: they classify Food Away from Home as non-tradable and exclude some transportation items (e.g., airline fare).

⁹We use data on CPI item “services less rent of shelter”.

where i , s , and t denote MSA, state, and month, respectively. $h = 1, \dots, H$ denotes the estimation horizon. We control for lags up to $K = 3$. α_t^h is a time fixed effect. η_i^h is an MSA fixed effect. We cluster the standard errors $u_{i,t}^h$ at the MSA level. Our inflation measure, $y_{i,t}$, is year-on-year services inflation (excluding housing) in MSA i in month t calculated as the logarithmic difference in the prices for services excluding housing, $\ln(\text{prices})_{i,t} - \ln(\text{prices})_{i,t-12}$. Hence, the dependent variable $y_{i,t}^h - y_{i,t}^0$ captures the change in inflation over a horizon of h months from time t to time $t+h$. $w_{i,t}$ is our independent variable of interest. It is year-on-year growth of average wages in the services sector from Homebase calculated as the logarithmic difference, $\ln(\text{wage})_{i,t} - \ln(\text{wage})_{i,t-12}$. We calculate average wages at the MSA-month level. For each worker in the Homebase dataset, we compute total monthly wages and hours worked, then derive their average hourly wage. These worker-level hourly wages are then averaged across all individuals employed in the same MSA and month. To limit the influence of outliers, we winsorize both wages and hours at the 1st and 99th percentiles within each month.

We include a vector of controls $X_{i,s,t}$ at the state-year level. This includes labor productivity (from BLS) as a proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline inflation and core inflation excluding food and energy, and residuals from regressions of previous horizons (i.e., from 1 to $h-1$). According to Teulings and Zubanov (2014), these residuals account for information between time t and $t+h$ that are not fully controlled by the other independent variables (which are as of time t).

The coefficient β^h captures the impulse response of inflation on wage growth over the horizon h . We refer to this coefficient as the wage-price pass-through. The causal interpretation of this coefficient rests on several grounds. Empirically, wages are typically less flexible than prices, making it unlikely that they will quickly adjust to changes in inflation. This is also evident in our sample period. As Figure 3 shows, the rise in wages precedes the rise in inflation.¹⁰

We instrument wage growth with a Bartik shock, following the intuition of Bartik (1991) and more recently Hazell et al. (2022).¹¹ As discussed in more detail below, this instrument is intended to isolate wage pressures arising from increased local demand in the service sector. This is important for addressing the confounding effects of simultaneous supply and demand shocks, which typically influence the slope of the Phillips curve in opposite directions. In a nutshell, the Bartik shock is a shift-share instrument constructed based on national industry-level labor market tightness in the tradable sector (shift) and the employment share of those tradable industries within each MSA (share).

A key innovation in our construction of the shift variable is that we use national changes in labor market tightness in the tradable sector only. Because prices in tradable sector are set in national or global markets, this measure is unlikely to be directly correlated with unobserved shocks to wage growth in local non-tradable (service) sectors. In other words, we assume that changes in labor market tightness in the tradable sector are driven by shocks that are specific to the tradable sector and uncorrelated with local supply shocks for

¹⁰The lagged response of inflation to wage growth is a common identification assumption in structural vector autoregression (SVAR) estimations. See, for example, Bernanke and Blanchard (2024b).

¹¹See Soh et al. (2022) and Chen and Lee, 2024 for applications of this method in the context of the pandemic.

non-tradables. Importantly, as positive (negative) shocks in the tradable sector increases (decreases) the demand for local non-tradables, firms in the non-tradable sector will experience an increase (decrease) in competition for workers, which ought to result in them raising (dropping) wages and prices.

Our primary measure of labor market tightness is the vacancy-to-unemployment ratio (i.e., the number of vacancies divided by the number of unemployed workers), following recent studies (e.g., Ball et al., 2022; Blanchard et al., 2022; Benigno and Eggertsson, 2023). Several researchers have argued that the vacancy-to-unemployment ratio better reflects labor market tightness in the post-pandemic economy given the upward shift in the Beveridge curve post-pandemic (Furman and Wilson (2021), Barnichon and Shapiro (2024), and Domash and Summers (2022)). The Beveridge curve, which shows the number of vacancies per unemployed worker, has shifted upward, suggesting a similar shift in the traditional unemployment-based Phillips curve, which implies higher inflation at any given unemployment rate. Therefore, using the vacancy-to-unemployment ratio better captures this dynamic. That said, some researchers believe that movements in vacancies may be endogenous to inflation (Afrouzi et al., 2024; Bagga et al., 2025; Guerreiro et al., 2024; and Pilossoph and Ryngaert, 2024). Therefore, we also employ unemployment rate, employment growth, or wage growth to construct alternative instruments.

Formally, the Bartik shock based on the vacancy-to-unemployment ratio in MSA i in month t is defined:

$$Shock_{i,t} = \widehat{\ln(V)}_{i,t} - \widehat{\ln(U)}_{i,t}, \quad (2)$$

where $\widehat{\ln(V)}_{i,t}$ is calculated as

$$\widehat{\ln(V)}_{i,t} = \sum_{j=1}^J \phi_{i,j,2015-2016} * \ln(V_{j,t}). \quad (3)$$

$\ln(V_{j,t})$ is the log number of vacancies for 3-digit NAICS industry j in time t . $\phi_{i,j,2015-2016}$ is the average employment share of industry j in MSA i in 2015-2016. The set of industries is restricted to those classified as part of the tradable sector, following the definition in Hazell et al. (2022). We calculate $\widehat{\ln(U)}_{i,t}$ similarly. One way to think about the shock, $Shock_{i,t}$, is that it uses labor market tightness as a proxy for latent quasi-experimental demand shocks to the local market—a point we will revisit in detail below.

We use industry-level data on unemployment from the CPS and vacancy data from a proprietary dataset provided by Indeed. Indeed is a global search engine for job listings that collects job postings from various sources, including job listing sites, employer career sites, and applicant tracking systems. Duplicated listings are removed, so each job is shown only once, even if it appears on multiple platforms. Our final dataset includes 142 million job postings, covering 421 occupations based on ISCO-08 classifications across 2.9 million companies and 576 counties in the US.

The advantage of using Indeed, as opposed to the Job Openings and Labor Turnover Survey (JOLTS) from the BLS, is its more granular industry classifications, which enable us to capture shocks at the 3-digit NAICS level. A limitation of the Indeed dataset is that it may not capture all job vacancies, as some are not posted online. To address this, we rescale the vacancies from Indeed into nationally representative units, and we describe this adjustment

procedure in Appendix A.3. Additionally, Indeed allows us to compute the vacancy-to-unemployment ratio at the MSA-month level—a measure not available from JOLTS. We use this measure to classify local labor markets with labor shortages when investigating the nonlinearity of price-wage pass-through based on local labor market conditions.

The identification assumption underlying the Bartik shock, as Borusyak et al. (2022) show, is that demand shocks to the local market are quasi-randomly assigned. This means that the shocks are uncorrelated with relevant unobservables in expectation and that a shock-level law of large numbers applies. Several points warrant discussion regarding this identification assumption.

First, the demand shocks are assumed to be orthogonal to unobserved supply shocks in local markets. This orthogonality condition is satisfied in our case by using national variations in the tradable sector, under the assumption that prices in tradable industries are set at the national or international level and are uncorrelated with local market conditions.

Second, when shocks are measured using equilibrium objects—such as our baseline vacancy-to-unemployment ratio and alternative measures of employment or wage growth—it can be challenging to view them as quasi-randomly assigned. In such cases, the Bartik shock can be interpreted as a noisy estimate of the latent true shock, with the estimator being asymptotically valid under certain conditions. One such condition is when the supply shocks are spatially uncorrelated, in which case a leave-one-out (LOO) shock estimator is valid.¹² In fact, in settings where many regions contribute to the shock calculation, even the conventional non-LOO estimator is asymptotically valid. Intuitively, this requires that the typical industry is in a much larger number of regions than the number of industries that a typical region specializes in, which holds in our setting. Furthermore, since our local markets are MSAs that are geographically dispersed across different regions, it is reasonable to assume that local labor supply shocks are not highly spatially correlated. To further mitigate potential spatial correlation, we also control for region $\times$ time fixed effects to account for time-varying regional conditions. As a preview of our results, we find that estimates from the LOO estimator closely align with those from the conventional estimator, and controlling for regional conditions yields similar findings. Section 5 provides further robustness checks with alternative specifications and additional controls.

Third, the law of large numbers assumption requires that the instrument incorporates many sufficiently independent shocks, each with a sufficiently small average exposure. Intuitively, this means including shocks from many industries relative to the number of local markets in the estimation. To satisfy this requirement, we calculate the shocks at the 3-digit NAICS industry level rather than at more aggregated industry levels.

Finally, a more subtle point arises from the exposure weights in the Bartik shock. Intuitively, this means that the demand shock for a tradable industry used to construct the instrument should be uncorrelated with supply shocks in the local markets most specialized in that industry. For example, the oil industry experiencing an oil boom shock should not face systematically different unobserved supply conditions (e.g., migration patterns or worker preferences) in primary oil production areas. Put differently, unobserved local supply shocks threaten identification only if they are systematically correlated with the demand shocks of the tradable industries in which a local market is most specialized. While there is

¹²The LOO estimator is calculated by leaving out observations from MSA i when calculating $\ln(V_{j,t})$ and $\ln(U_{j,t})$ for $Shock_{i,t}$.

no strong reason to expect such correlation, we address this potential concern by controlling for potential local supply shocks to headline inflation. These are defined at the MSA-month level as the difference between year-on-year headline inflation and core inflation excluding food and energy. Ball et al. (2022) show that headline inflation shocks during our sample period were mainly driven by supply-side factors such as fluctuations in energy prices and prices in auto-related industries. To the extent that headline inflation shocks vary across MSAs, they may capture the differing direct impacts of these supply-side factors across regions. Additionally, we control for labor productivity at the state-year level to reflect unit labor costs. In Section 5, we discuss the robustness of results with additional controls and other specifications.

5 Results

Baseline Table 1 presents the summary statistics of the sample. Figure 4 illustrates the baseline results for equation (1). The first stage estimates are positive and statistically significant across all horizons, indicating a positive pass-through from tight labor markets to wages. Similarly, the second stage estimates are positive and statistically significant for horizons ranging from 2 to 12 months, reflecting a positive pass-through from wages to prices.

The results are economically significant. Interpreting the logarithmic difference as a growth rate, the first stage estimate of 13.7 in the 3-month horizon implies that a one-log-point increase in the vacancy-to-unemployment ratio is associated with a 13.7-percentage-point increase in the year-on-year wage growth rate. This indicates a pass-through of 13.7 percent from the vacancy-to-unemployment ratio to wage growth. The effect remains stable over time, staying at 12.5 percent in the 10-month horizon.

For the second stage, the estimate of 30.3 percent over the 3-month horizon suggests that a one-percentage-point increase in year-on-year wage growth is associated with a 0.303-percentage-point increase in year-on-year inflation. This effect also increases over time, reaching 48.4 percent in the 10-month horizon.

Combining the two stages, a one-log-point increase in the vacancy-to-unemployment ratio corresponds to an increase in year-on-year inflation of 4.1 percentage points over the 3-month horizon, rising to 6.0 percentage points over the 10-month horizon.

We conduct a decomposition exercise for inflation drivers using the baseline estimation results. Figure 5 illustrates the decomposition results based on the 3-month estimations averaged across the MSAs. We find that local labor market tightness, captured by the log vacancy-to-unemployment ratio, emerged as the key driver of inflation since mid-2021. It accounted for, on average, 39.8 percent of the inflation between Q3 2022 and Q1 2023. This result contrasts with the pre-pandemic period and the post-pandemic period up to mid-2021, when common shocks—captured by the time fixed effects—largely offset the effects of local labor market tightness.

These results hold across all MSAs in the sample. Figure 6 presents the decomposition exercise at the MSA level. While the timing and magnitude of the inflation surge vary across MSAs, the model accurately predicts actual inflation in each MSA. Local labor market tightness remains a key driver of inflation between Q3 2022 and Q1 2023 in all

MSAs. The share of inflation explained by local labor market tightness ranges from 20.4 to 33.9 percent, with an average of 26.3 percent (see Figure 7).

Extension and robustness Recent theoretical and empirical work has emphasized the importance of nonlinearity in Phillips curves (Ball et al., 2022; Benigno and Eggertsson, 2023; Dao et al., 2024). To explore potential nonlinearity in our sample, we first examine whether the estimated coefficients vary with the extent of labor shortages. As defined in Benigno and Eggertsson (2023), a labor shortage occurs when the vacancy-to-unemployment ratio exceeds 1, indicating more job vacancies than workers seeking employment.

Figure 8 shows the results when we split the sample into two subsamples based on whether the MSA faced a labor shortage (“High V/U”) or not (“Low V/U”). We find that the first stage results are similar across both subsamples. However, the second stage results are larger in the presence of a labor shortage, particularly over the 6- to 12-month horizon. In results not reported here, we also explore nonlinearity by adding quadratic and cubic terms of the log vacancy-to-unemployment ratio to the baseline regression. While the estimated quadratic term is negative and the cubic term is positive, neither is statistically significant.

As discussed in 4, the validity of our baseline estimator relies on assumptions of spatially uncorrelated unobserved local supply shocks and a sufficiently large number of regions contributing to the shock calculation. We now investigate the robustness of our results when these assumptions might be violated. One approach to address potential biases from regions contributing to their own shocks is to employ a leave-one-out (LOO) estimator. Figure 9 shows that results from a LOO estimator are similar to the baseline. To assess robustness against potential spatial correlation, we include controls at the regional level. Since our local markets are MSAs that are geographically dispersed, spatial correlation is most likely to occur among MSAs within the same region. Accordingly, we control for time-varying economic conditions at the regional level using region $\times$ time fixed effects. As shown in Figure 10, our results remain robust with these controls. Our next approach aims to directly address potential confounding factors at the MSA level, such as rising household income and housing wealth during our sample period—possibly reflecting stimulus spending and rising housing prices—that could affect local labor supply. In Figure 11, we additionally control for household income and housing wealth at the MSA level. Our results are robust to these controls.

Finally, we explore alternative measures of labor market tightness. In our baseline estimation, We use vacancy-to-unemployment ratio following a recent literature, which account for the upward shift in the Beveridge curve in the post-pandemic economy. Here, we examine how results based on traditional measures of labor market tightness compares, also addressing the potential concern that vacancies may reflect factors other than labor market tightness. Specifically, we use the unemployment rate, employment growth, or wage growth as the shift variable to construct the Bartik shock. The results are shown in Figure 12 for the unemployment rate measure, Figure 13 for the employment growth measure, and Figure 14 for the wage growth measure.

Comparison with recent literature How do our results compare to recent estimations that use aggregate data? Benigno and Eggertsson (2023) estimate the pass-through from

the log vacancy-to-unemployment ratio to core inflation (measured by all items excluding food and energy) over a 1-quarter horizon. They find a pass-through of 4.7 percentage points with labor shortage (i.e., when the vacancy-to-unemployment ratio is less than 1) and 0.5 percentage points without labor shortage in their 2008-2022 sample. In comparison, our results suggest a pass-through of 4.6 percentage points with labor shortage and 2.8 percentage points without labor shortage.

Ball et al. (2022) estimate the pass-through from the log vacancy-to-unemployment ratio to core inflation (measured by median inflation) and uncover nonlinearity with respect to the vacancy-to-unemployment ratio. For example, their results imply that an increase in the vacancy-to-unemployment ratio from 0.5 to 1.5 leads to a 3.5 percentage point rise in core inflation. In contrast, our results indicate a 4.6 percentage point increase in services inflation (excluding housing) for a similar change in the vacancy-to-unemployment ratio.

Overall, our results are comparable to, although slightly larger than, recent estimates based on aggregate data. Differences in the estimates can be attributed to several factors, including variations in data and methods. We expect the pass-through of labor market tightness to be stronger for services inflation compared to core inflation, as services are typically more labor-intensive and less tradable than goods. Additionally, as Hazell et al. (2022) demonstrate in a theoretical model, the slope of the regional Phillips curve is steeper than that of the aggregate Phillips curve when labor market conditions are persistent.¹³ As we discuss below, this difference between cross-sectional and aggregate Phillips curves is also observed empirically.

External validity Our analysis relies on wage data from Homebase, which provides more granular information than official statistics. This granularity is crucial because official data lack the spatial variation needed to estimate the Phillips curve at the MSA level (e.g., CPS and CES) or the high frequency required to capture rapid movements in post-pandemic wages (e.g., QCEW). To ensure the robustness of our results, we conduct a series of tests to evaluate the external validity of our results.

First, we assess the external validity of the wage data from Homebase by comparing its wage trends and wage growth rates with official statistics. At the national level, Figure 15 shows that the average earnings trend from the Homebase sample closely aligns with those from the CPS and the CES in the services sector, corroborating similar findings in Dvorkin and Isaacson, 2022 and Chen and Lee, 2024. Figure 16 shows that year-on-year hourly wage growth from the Homebase sample also close track that from the CES in similar service industries. At the state level, the monthly wage changes from Homebase and the CES are also highly correlated as shown in Chen and Lee, 2024.

We then assess the external validity of our results on wage-price pass-through by replicating our results with official statistics with lower frequency. Since no official wage data are available at the MSA-month level, we use MSA-quarter level data from the QCEW. The sample period for this estimation is from Q1 2019 to Q4 2022 as the baseline.

As shown in Figure 17, the first stage coefficient ranges from 30.2 to 43.6 over the 1- to 4-quarter horizon. The second stage coefficient increases over the same horizon, ranging

¹³The estimation of an aggregate Phillips curve is also influenced by how long-run inflation expectations are measured. In the regional Phillips curve within a monetary union, long-run inflation expectations are captured by time fixed effects.

from 0.14 to 0.16. The combined estimations from both stages suggest a pass-through that peaks at 6.0 percentage points in the 3-quarter horizon. While these magnitudes are similar to those in the baseline, the coefficients are less precisely estimated with the QCEW sample. Figure 18 decomposes the estimation results at the 1-quarter horizon. We again find that the model closely predicts actual inflation and that local labor market tightness has emerged as the key driver since mid-2021. Overall, our findings on wage-price pass-through from the Homebase data are broadly consistent with evidence from official wage data, despite the limitations of the latter in terms of aggregation and frequency.

Finally, we examine the external validity of our findings regarding the role of labor market tightness in the time series. This exercise aims to place our results within the context of a longer time series and to compare them with the literature on the aggregate Phillips curve. To achieve this, we adopt a standard form of the aggregate Phillips curve with labor market tightness measured by the log vacancy-to-unemployment ratio. Details of the model and data are discussed in Appendix A.2, and key findings are summarized here.

As shown in Figure 20, we find a significantly positive coefficient for the log vacancy-to-unemployment ratio, indicating an upward-sloping Phillips curve. The coefficient peaks between 12 and 24 months before declining. Specifically, it is 0.6 at the 3-month horizon and increases to between 1.5 and 2.0 at the 12-month horizon and beyond. Using rolling regressions with a 3-year window, we observe substantial time variation in the coefficient.¹⁴

Another way to highlight the important role of labor market conditions in post-pandemic services inflation (excluding housing) is to compare models with and without account for labor market condition. Figure 21 shows that the fitted values from the rolling window regression (“Fitted” line) with labor market tightness closely align with actual inflation outcomes at the national level, effectively tracking the large surge and subsequent decline in inflation during 2021-2023. In contrast, a model that does not incorporate labor market dynamics (by imposing that V/U equals 1, as shown by the “Fitted w/o $\ln(V/U)$ ” line) predicts an earlier inflation surge and a quicker fall in inflation. Under this model, peak inflation would have been 1 percentage point lower, and inflation would have fallen below 2 percent by early 2023—contrary to the observed data.

Another way to examine the broader relevance of our results is to compare the model’s fitted values with actual inflation outcomes. In this regard, we have shown evidence at the MSA level using our baseline model in Figure 7) as well as a time series regression at the national level. Both approaches confirm that the fitted values closely align with actual inflation outcomes.

In summary, we show that wage trends in the Homebase sample closely track official statistics. Moreover, replicating the Homebase sample using official data—though with simplified specifications of lower frequency, less granularity, or without directly estimating wage-price pass-through—yields corroborating evidence on the role of labor market tightness in driving services inflation. Furthermore, our model’s fitted values closely align with actual inflation outcomes. Taken together, these results reinforce the external validity of the Homebase data, suggesting that our results have broader relevance for understanding wage-inflation dynamics in the U.S. service sector as a whole. While alternative data sources

¹⁴Benigno and Eggertsson (2023) report a coefficient of 0.6 in the full sample and 4.5 for the post-2008 sample during periods of labor shortage.

provide supporting evidence, the Homebase data—with its cross-sectional granularity and high time frequency—enables accurate estimation of the wage-price pass-through, something other data cannot achieve.

6 Conclusion

We use a proprietary micro dataset on wages to estimate wage-price pass-through in the United States, exploiting variation in labor market tightness across MSAs. Our findings reveal that service sector wage growth plays an important role in services inflation through the local labor market tightness channel. The pass-through from a tight labor market to service sector wage growth, and from service sector wage growth to services inflation, are both strong. Taken together, local labor market tightness emerged as a key driver of inflation between Q3 2022 and Q1 2023. These results suggest that the effects of overheated labor markets can be persistent when labor market tightness endures, and that significant wage growth may impede efforts to curb inflation.

	Obs	Mean	Std. Dev.	Min	Median	Max
Services inflation (excl. housing)	906	3.30	2.41	-3.18	2.87	10.03
Wage growth	906	0.95	17.22	-66.33	3.40	63.33
Bartik shock	906	-0.05	0.64	-2.05	0.02	0.83
Vacancy-to-unemployment ratio	906	1.07	0.54	0.15	1.04	3.53
Labor productivity	906	1.94	3.10	-4.50	2.10	7.70
Headline shock	906	0.54	1.03	-1.81	0.35	3.89
Log household income per capita	906	11.15	0.19	10.65	11.13	11.73
Log housing wealth	906	4.97	0.23	4.40	4.97	5.55

Table 1: Summary statistics

Notes: This table reports the summary statistics. The sample is a monthly panel of MSAs over January 2019 to December 2022. Observations are weighted by the average size of the labor force in each MSA at 2019. We measure inflation as year-on-year logarithmic difference in the price level, $\ln(prices)_{i,t} - \ln(prices)_{i,t-12}$ for MSA i and month t . Wage growth denotes year-on-year wage growth in the service sector from Homebase measured by logarithmic difference, $\ln(wage)_{i,t} - \ln(wage)_{i,t-12}$. We instrument wage growth using the Bartik shock for defined in equation (2) and constructed using tradable-sector industries following the definition in Hazell et al. (2022). Labor productivity (from BLS) proxies for local labor demand. Headline inflation shocks are defined as the difference between year-on-year headline and core excluding food and energy inflation. Housing wealth is defined as the product between Census homeownership rates and the Freddie Mac House Price Index. Source: BLS, Homebase, and authors' calculations.



Figure 1: Headline inflation

Notes: This figure plots the year-on-year U.S. headline inflation. Source: BLS and authors' calculations.

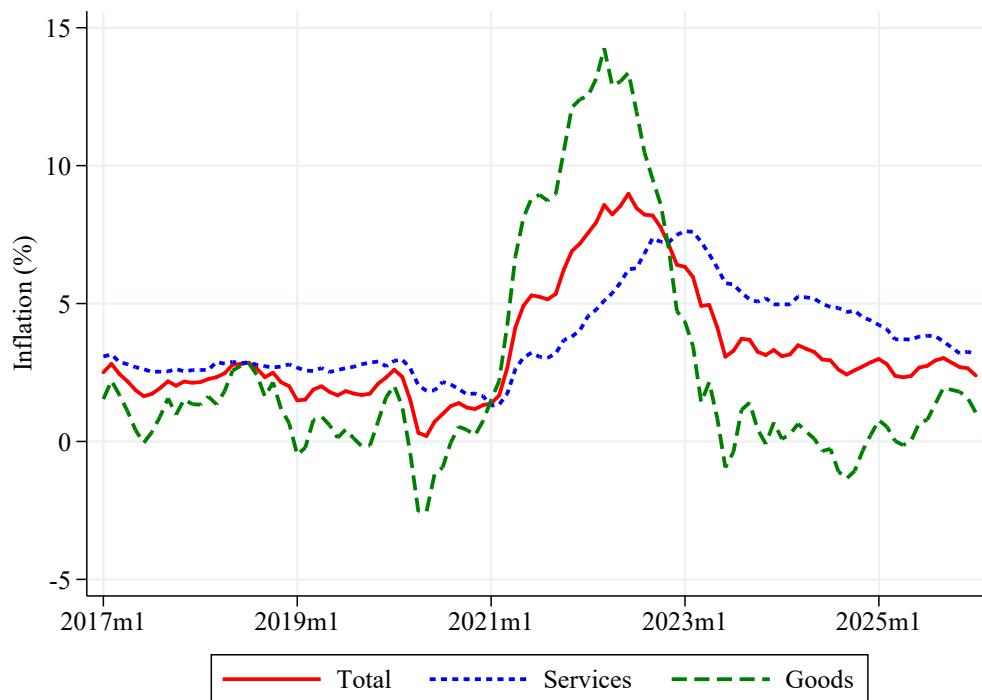


Figure 2: Services and goods inflation

Notes: This figure plots the year-on-year CPI inflation for all items, services, and goods. Source: BLS and authors' calculations.

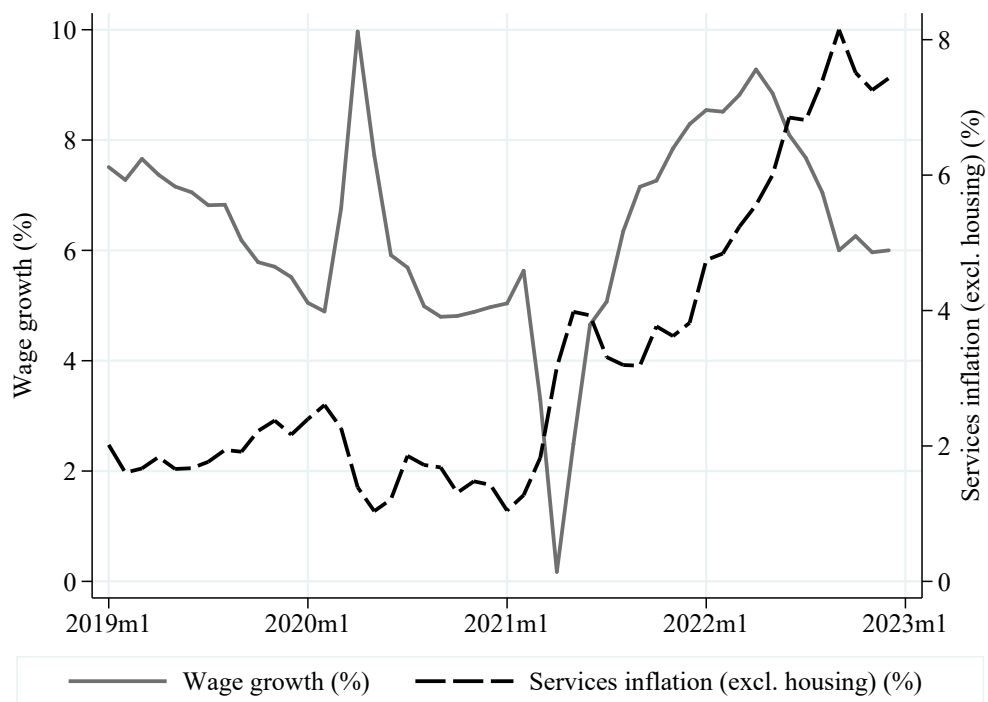


Figure 3: Price and wage inflation

Notes: This figure plots year-on-year average hourly wage growth from Homebase (left axis) and year-on-year services (excluding housing) inflation (right axis) from the BLS. Source: BLS, Homebase, and authors' calculations.

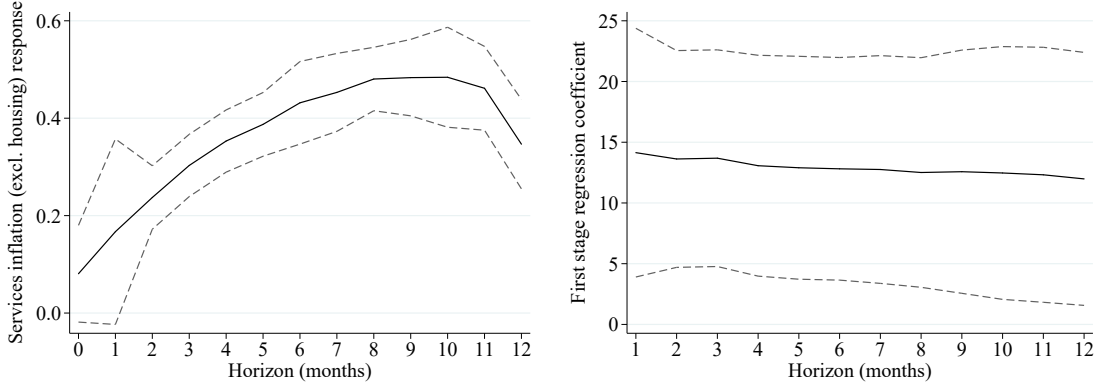


Figure 4: Services inflation (excluding housing) response, MSA level baseline model

Notes: This figure plots first and second stage estimates from the LP-IV model in equation (1):

$$y_{i,t}^h - y_{i,t}^0 = \alpha_t^h + \eta_i^h + \beta^h w_{i,t} + \sum_{k=1}^K \gamma_k^h w_{i,t-k} + \sum_{k=0}^K \delta_k^h y_{i,t-k} + \gamma^h X_{i,s,t} + u_{i,t}^h$$

where i , s , and t denote MSA, state, and month respectively. $h = 1, \dots, H$ denotes the estimation horizon. We control for lags up to $K = 3$. $y_{i,t}$ is year-on-year inflation of MSA i in month t measured by the logarithmic difference in the prices for services excluding housing, $\ln(prices)_{i,t} - \ln(prices)_{i,t-12}$. The dependent variable $y_{i,t}^h - y_{i,t}^0$ captures the difference in inflation over a horizon of h months from time t to time $t + h$. α_t^h is a time fixed effect. η_i^h is an MSA fixed effect. $w_{i,t}$ is our independent variable of interest, year-on-year wage growth in the service sector from Homebase measured by logarithmic difference, $\ln(wage)_{i,t} - \ln(wage)_{i,t-12}$. We instrument wage growth using the Bartik shock defined in equation (2) and constructed using tradable-sector industries following the definition in Hazell et al. (2022). We also include a vector of controls $X_{i,s,t}$ at the state-year level, including labor productivity (from BLS) to proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline and core excluding food and energy inflation, and residuals from the regressions for previous horizons. The sample is a monthly panel of MSAs over January 2019 to December 2022. Observations are weighted by the average size of the labor force in each MSA at 2019. We cluster the standard errors $u_{i,t}^h$ at the MSA level. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Source: BLS, Homebase, and authors' calculations.

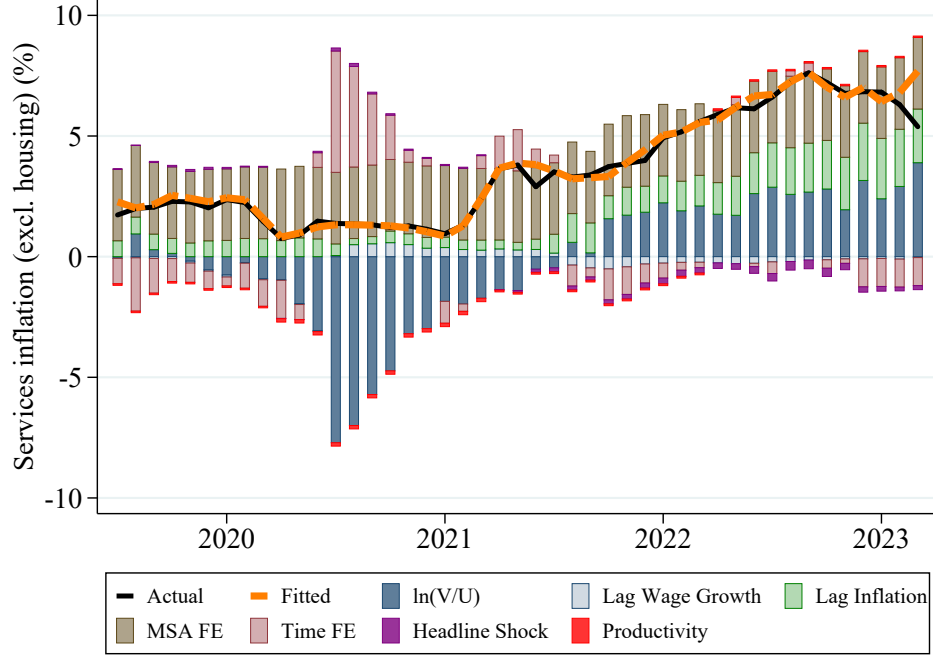


Figure 5: Predictions for services inflation (excluding housing), MSA level baseline model

Notes: This figure plots fitted values from the LP-IV model estimated at the MSA level (Figure 4):

$$y_{i,t}^h = \alpha_{2,t}^h + \eta_{2,i}^h + \beta_2^h \widehat{w}_{i,t} + \sum_{k=1}^K \gamma_{2,k}^h w_{i,t-k} + \sum_{k=0}^K \delta_{2,k}^h y_{i,t-k} + \gamma_2^h X_{i,s,t} + u_{2,i,t}^h$$

$$w_{i,t} = \alpha_{1,t}^h + \eta_{1,i}^h + \beta_1^h Shock_{i,t} + \sum_{k=1}^K \gamma_{1,k}^h w_{i,t-k} + \sum_{k=0}^K \delta_{1,k}^h y_{i,t-k} + \gamma_1^h X_{i,s,t} + u_{1,i,t}^h$$

where i , s , and t denote MSA, state, and month respectively. $h = 3$ denotes the estimation horizon. All values are unweighted averages across MSAs in the sample. The solid line plots actual inflation, and the dashed line plots fitted values from the LP-IV model. The bars show the contribution of each independent variable in the LP-IV model, where the contribution captures the combined effect of each variable from the first and second stages. “ln(V/U)” plots the fitted value from the Bartik shock $Shock_{i,t}$ for the log vacancy-to-unemployment ratio defined in equation (2), constructed using only tradable-sector industries following the definition in Hazell et al. (2022). “Lag Wage Growth” contains fitted values from $w_{i,t-k}$, “Lag Inflation” from $y_{i,t-k}$, “MSA FE” from $\alpha_{1,t}^h$ and $\alpha_{2,t}^h$, “Time FE” from $\eta_{1,t}^h$ and $\eta_{2,t}^h$, and “Headline Shock” and “Productivity” from the controls in $X_{i,s,t}$. Source: BLS, Homebase, and authors’ calculations.

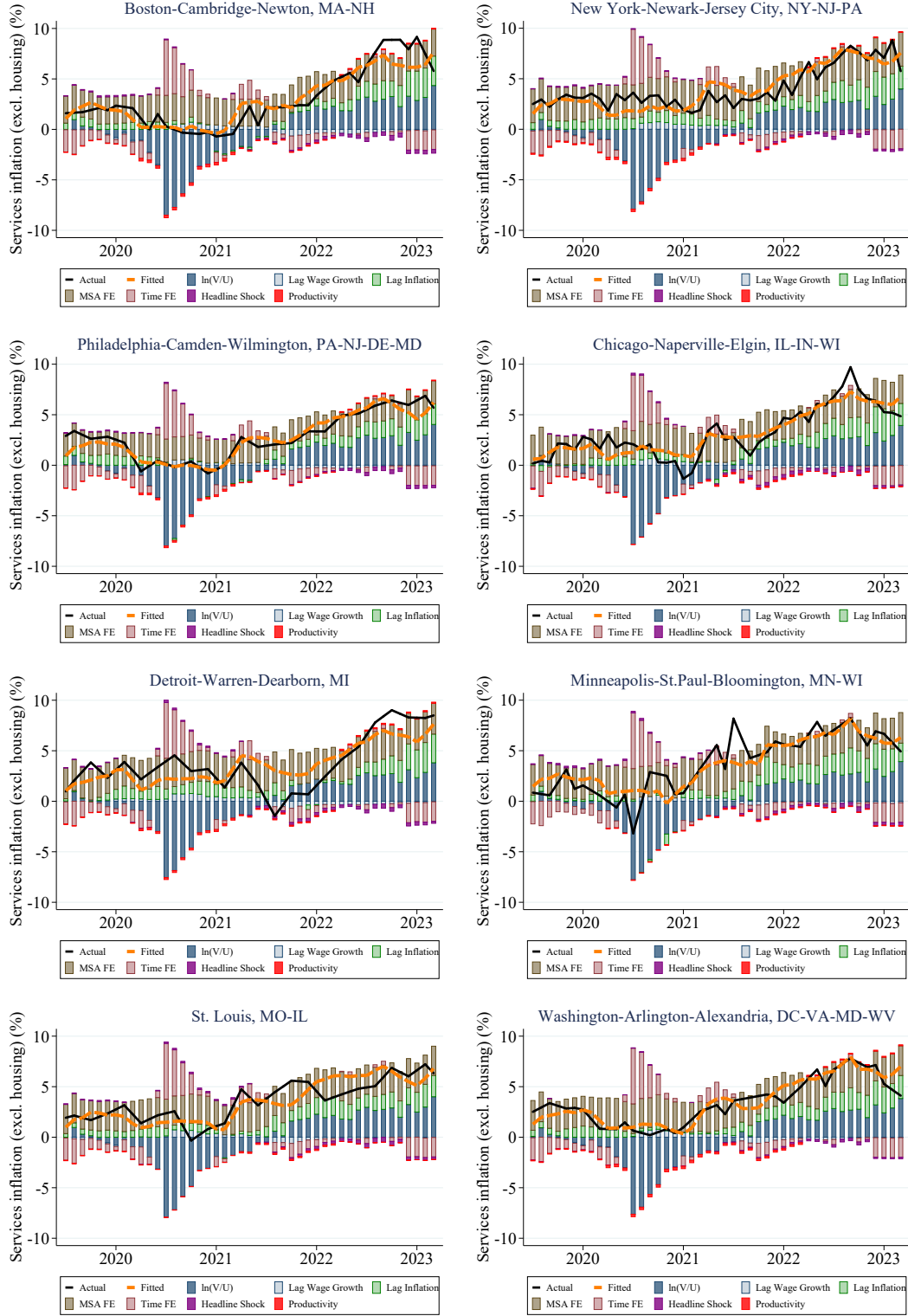


Figure 6: Predictions for services inflation (excluding housing) by MSA

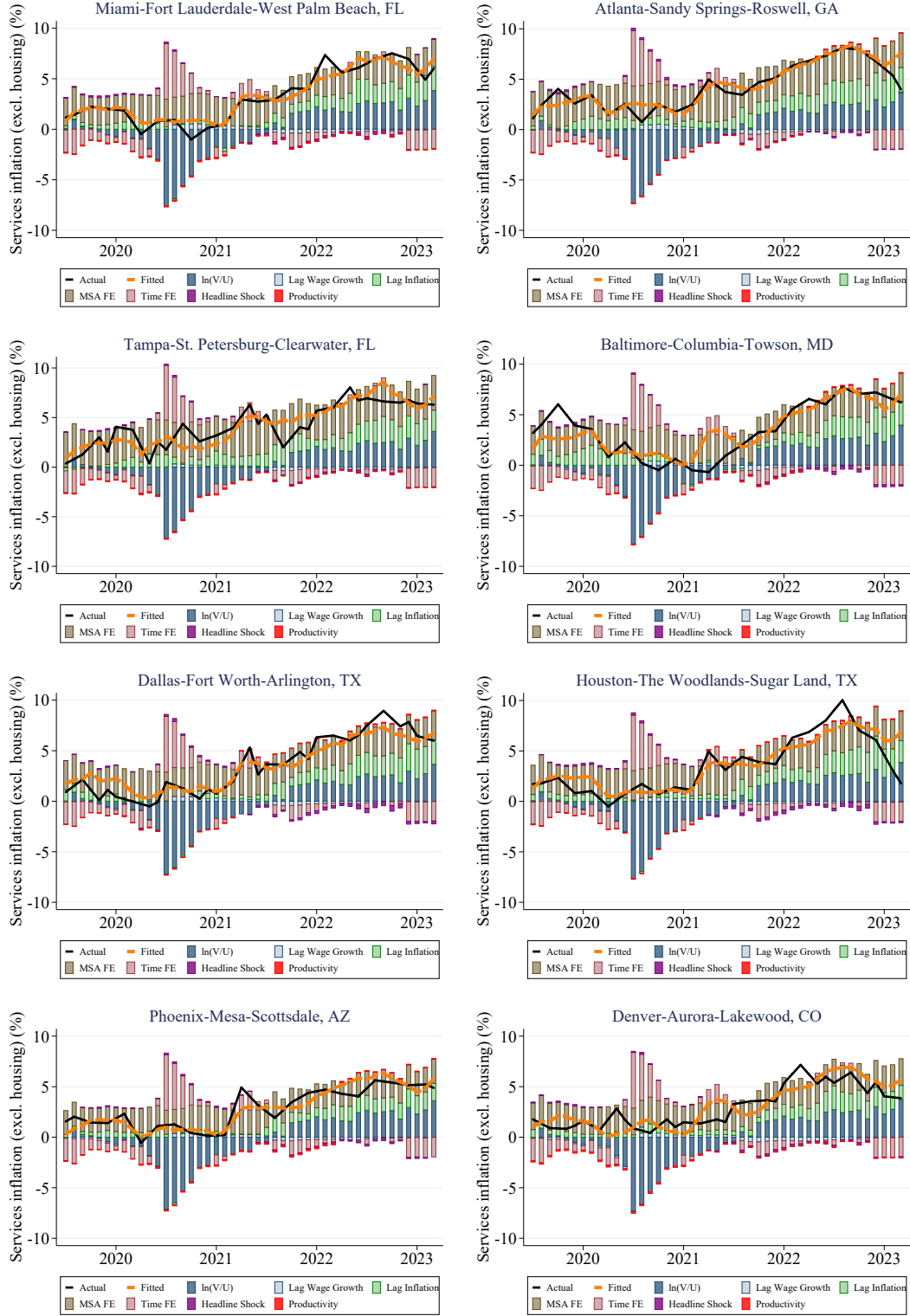


Figure 6: Predictions for services inflation (excluding housing) by MSA (Continued)

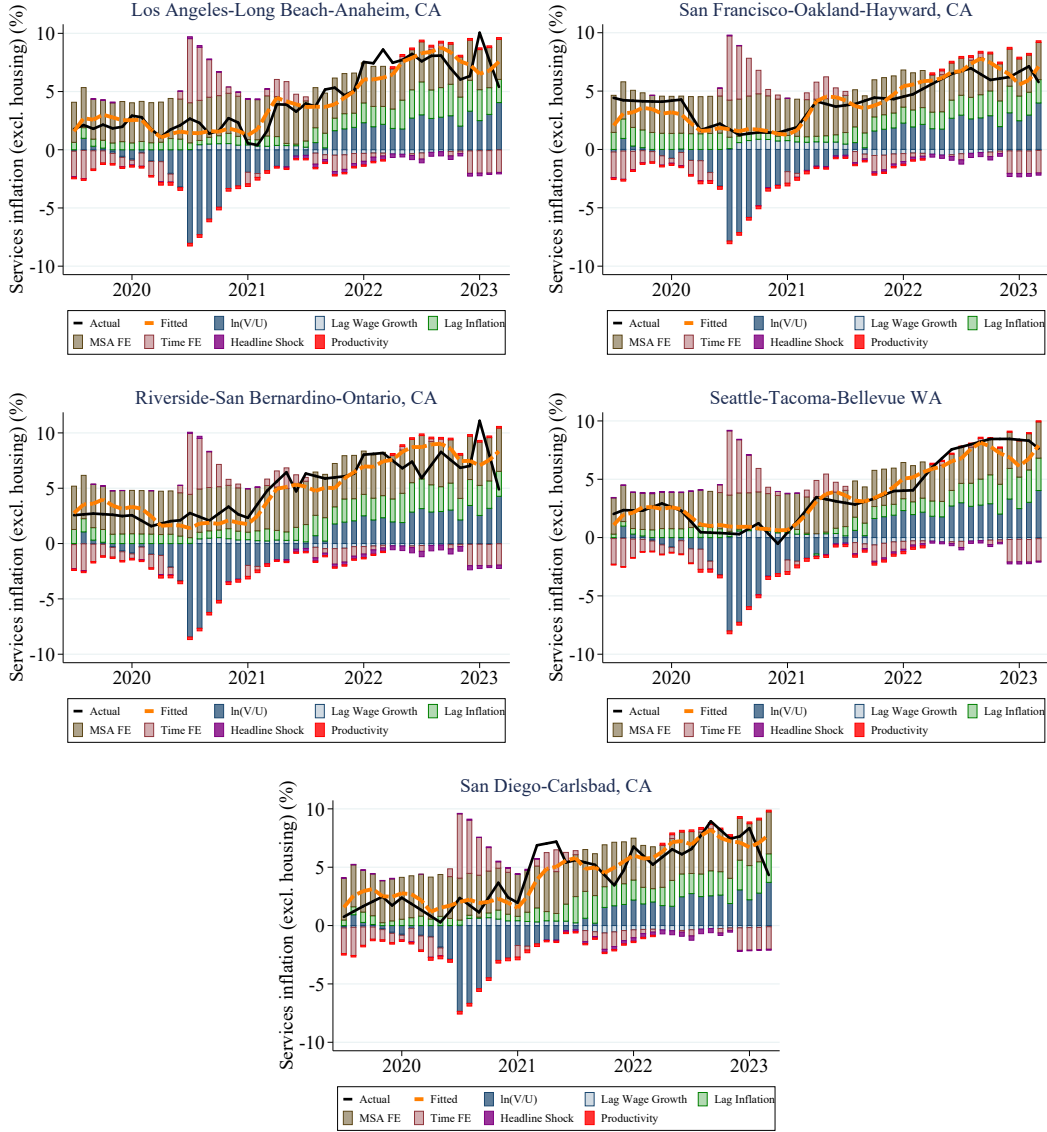


Figure 6: Predictions for services inflation (excluding housing) by MSA (Continued)

Notes: This figure plots fitted values from the LP-IV model estimated at the MSA level (Figure 4). The solid line plots actual inflation, and the dashed line plots fitted values from the LP-IV model. The bars show the contribution of each independent variable in the LP-IV model, where the contribution captures the combined effect of each variable from the first and second stages. “ln(V/U)” plots the fitted value from the Bartik shock $Shock_{i,t}$ defined in equation (2) and constructed using tradable-sector industries following the definition in Hazell et al. (2022). “Lag Wage Growth” contains fitted values from $w_{i,t-k}$, “Lag Inflation” from $y_{i,t-k}$, “MSA FE” from $\alpha_{1,t}^h$ and $\alpha_{2,t}^h$, “Time FE” from $\eta_{1,t}^h$ and $\eta_{2,t}^h$, and “Headline Shock” and “Productivity” from the controls in $X_{i,s,t}$. Source: BLS, Homebase, and authors’ calculations.

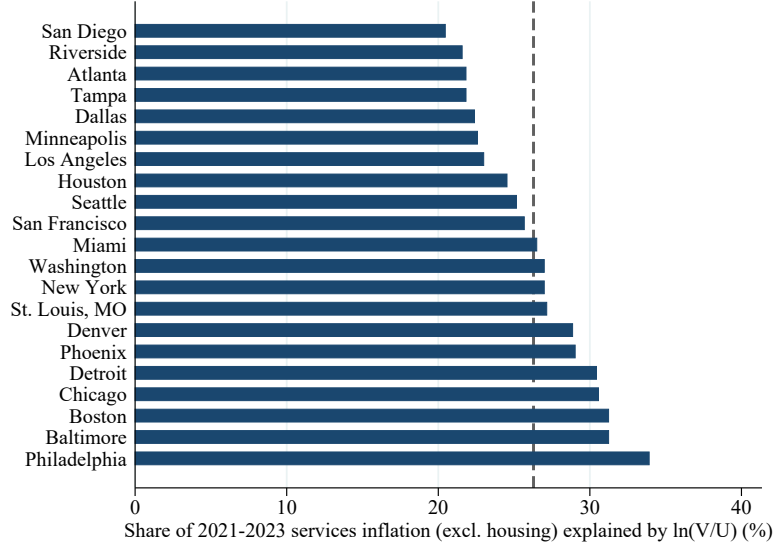


Figure 7: Share of services inflation (excluding housing) explained by $\ln(V/U)$

Notes: This figure plots the average share of inflation explained by $\ln(V/U)$ based on fitted values of the LP-IV model estimated at the MSA level at the 3-month horizon (Figure 4):

$$y_{i,t}^h = \alpha_{2,t}^h + \eta_{2,i}^h + \beta_2^h \widehat{w}_{i,t} + \sum_{k=1}^K \gamma_{2,k}^h w_{i,t-k} + \sum_{k=0}^K \delta_{2,k}^h y_{i,t-k} + \gamma_2^h X_{i,s,t} + u_{2,i,t}^h$$

$$w_{i,t} = \alpha_{1,t}^h + \eta_{1,i}^h + \beta_1^h Shock_{i,t} + \sum_{k=1}^K \gamma_{1,k}^h w_{i,t-k} + \sum_{k=0}^K \delta_{1,k}^h y_{i,t-k} + \gamma_1^h X_{i,s,t} + u_{1,i,t}^h$$

where i , s , and t denote MSA, state, and month respectively. $h = 3$ denotes the estimation horizon. Share of services inflation (excluding housing) explained by $\ln(V/U)$ is calculated at the MSA level as the average fitted value for $\ln(V/U)$ over January 2021 to March 2023, as a share of the average actual services inflation (excluding housing) over the same period. Black dashed line shows the average across MSAs. The fitted value for $\ln(V/U)$ refers to the Bartik shock $Shock_{i,t}$ defined in equation (2) and constructed using tradable-sector industries following the definition in Hazell et al. (2022). Source: BLS, Homebase, and authors' calculations.

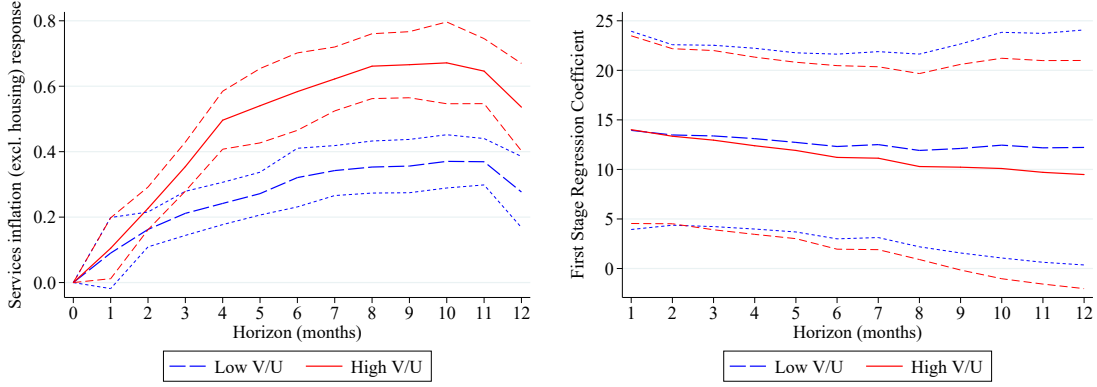


Figure 8: Services inflation (excluding housing) response, labor shortage subsample

Notes: This figure plots first and second stage estimates from the LP-IV model in equation (1) estimated over subsamples split by whether the MSA faced a labor shortage (“High V/U”) or otherwise (“Low V/U”):

$$y_{i,t}^h - y_{i,t}^0 = \alpha_t^h + \eta_i^h + \beta^h w_{i,t} + \sum_{k=1}^K \gamma_k^h w_{i,t-k} + \sum_{k=0}^K \delta_k^h y_{i,t-k} + \gamma^h X_{i,s,t} + u_{i,t}^h$$

where i , s , and t denote MSA, state, and month respectively. A labor shortage refers to a situation where the vacancy-to-unemployment ratio $V_{i,t}/U_{i,t}$ is greater than 1. $h = 1, \dots, H$ denotes the estimation horizon. We control for lags up to $K = 3$. $y_{i,t}$ is year-on-year inflation of MSA i in month t measured by the logarithmic difference in the prices for services excluding housing, $\ln(\text{prices})_{i,t} - \ln(\text{prices})_{i,t-12}$. The dependent variable $y_{i,t}^h - y_{i,t}^0$ captures the difference in inflation over a horizon of h months from time t to time $t + h$. α_t^h is a time fixed effect. η_i^h is an MSA fixed effect. $w_{i,t}$ is our independent variable of interest, year-on-year wage growth in the service sector from Homebase measured by logarithmic difference, $\ln(\text{wage})_{i,t} - \ln(\text{wage})_{i,t-12}$. We instrument wage growth using the Bartik shock defined in equation (2) and constructed using tradable-sector industries following the definition in Hazell et al. (2022). We also include a vector of controls $X_{i,s,t}$ at the state-year level, including labor productivity (from BLS) to proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline and core excluding food and energy inflation, and residuals from the regressions for previous horizons. The sample is a monthly panel of MSAs over January 2019 to December 2022. Observations are weighted by the average size of the labor force in each MSA at 2019. We cluster the standard errors $u_{i,t}^h$ at the MSA level. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Source: BLS, Homebase, and authors’ calculations.

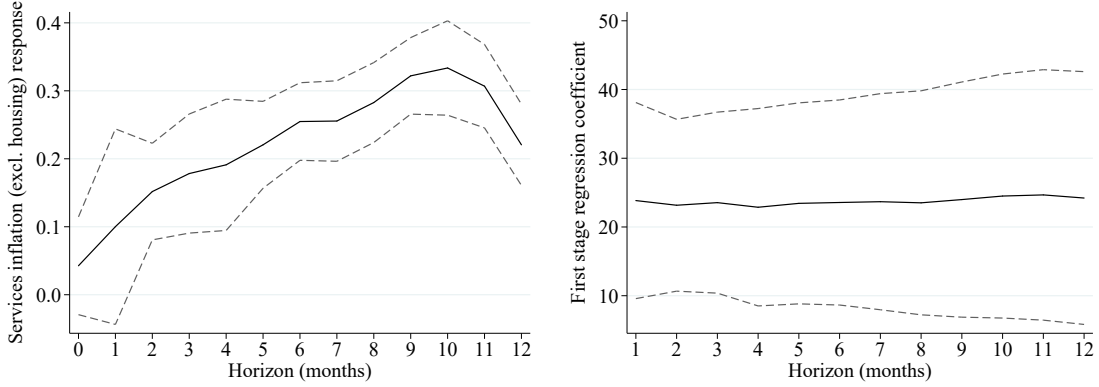


Figure 9: Services inflation (excluding housing) response, leave-one-out (LOO) estimator

Notes: This figure plots first and second stage estimates from the LP-IV model in equation (1):

$$y_{i,t}^h - y_{i,t}^0 = \alpha_t^h + \eta_i^h + \beta^h w_{i,t} + \sum_{k=1}^K \gamma_k^h w_{i,t-k} + \sum_{k=0}^K \delta_k^h y_{i,t-k} + \gamma^h X_{i,s,t} + u_{i,t}^h$$

where i , s , and t denote MSA, state, and month respectively. $h = 1, \dots, H$ denotes the estimation horizon. We control for lags up to $K = 3$. $y_{i,t}$ is year-on-year inflation of MSA i in month t measured by the logarithmic difference in the prices for services excluding housing, $\ln(prices)_{i,t} - \ln(prices)_{i,t-12}$. The dependent variable $y_{i,t}^h - y_{i,t}^0$ captures the difference in inflation over a horizon of h months from time t to time $t + h$. α_t^h is a time fixed effect. η_i^h is an MSA fixed effect. $w_{i,t}$ is our independent variable of interest, year-on-year wage growth in the service sector from Homebase measured by logarithmic difference, $\ln(wage)_{i,t} - \ln(wage)_{i,t-12}$. We instrument wage growth using the Bartik shock defined in equation (2) and constructed using a leave-one-out (LOO) approach following Borusyak et al. (2022). For each MSA, the national industry shock is computed excluding that MSA's own contribution. We also include a vector of controls $X_{i,s,t}$ at the state-year level, including labor productivity (from BLS) to proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline and core excluding food and energy inflation, and residuals from the regressions for previous horizons. The sample is a monthly panel of MSAs over January 2019 to December 2022. Observations are weighted by the average size of the labor force in each MSA at 2019. We cluster the standard errors $u_{i,t}^h$ at the MSA level. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Source: BLS, Homebase, and authors' calculations.

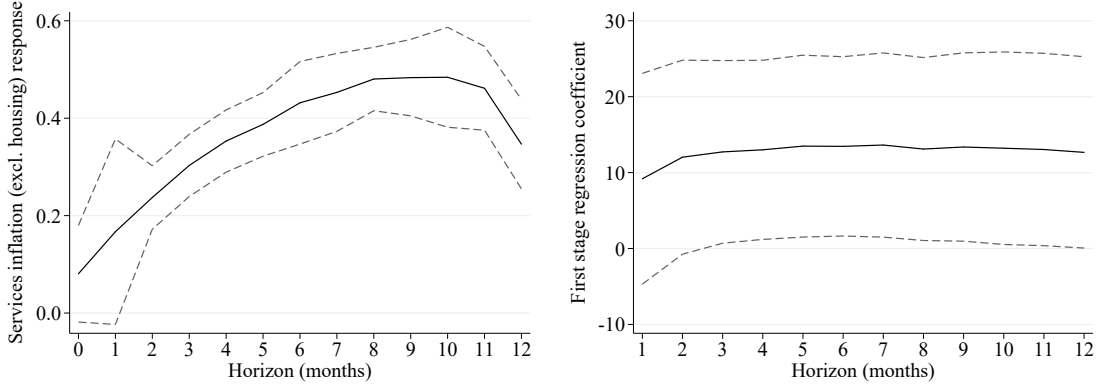


Figure 10: Services inflation (excluding housing) response, MSA level model with region-time fixed effects

Notes: This figure plots first and second stage estimates from the following LP-IV model:

$$y_{i,t}^h - y_{i,t}^0 = \alpha_{r,t}^h + \eta_i^h + \beta^h w_{i,t} + \sum_{k=1}^K \gamma_k^h w_{i,t-k} + \sum_{k=0}^K \delta_k^h y_{i,t-k} + \gamma^h X_{i,s,t} + u_{i,t}^h$$

where i , s , r , and t denote MSA, state, region, and month respectively. $h = 1, \dots, H$ denotes the estimation horizon. We control for lags up to $K = 3$. $y_{i,t}$ is year-on-year inflation of MSA i in month t measured by the logarithmic difference in the prices for services excluding housing, $\ln(\text{prices})_{i,t} - \ln(\text{prices})_{i,t-12}$. The dependent variable $y_{i,t}^h - y_{i,t}^0$ captures the difference in inflation over a horizon of h months from time t to time $t + h$. $\alpha_{r,t}^h$ is a region-time fixed effect. η_i^h is an MSA fixed effect. $w_{i,t}$ is our independent variable of interest, year-on-year wage growth in the service sector from Homebase measured by logarithmic difference, $\ln(\text{wage})_{i,t} - \ln(\text{wage})_{i,t-12}$. We instrument wage growth using the Bartik shock defined in equation (2) and constructed using tradable-sector industries following the definition in Hazell et al. (2022). We also include a vector of controls $X_{i,s,t}$ at the state-year level, including labor productivity (from BLS) to proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline and core excluding food and energy inflation, and residuals from the regressions for previous horizons. The sample is a monthly panel of MSAs over January 2019 to December 2022. MSAs are mapped to states based on the principal state on which the MSA belongs to. States are categorized into four Census Bureau designated regions. Observations are weighted by the average size of the labor force in each MSA at 2019. We cluster the standard errors $u_{i,t}^h$ at the MSA level. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Source: BLS, Homebase, and authors' calculations.

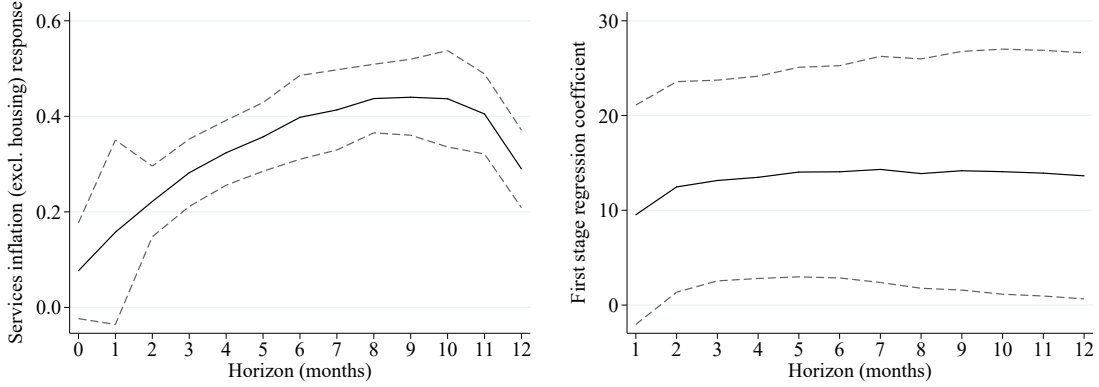


Figure 11: Services inflation (excluding housing) response, MSA level model with region-time fixed effects, household income, and housing wealth

Notes: This figure plots first and second stage estimates from the following LP-IV model:

$$y_{i,t}^h - y_{i,t}^0 = \alpha_{r,t}^h + \eta_i^h + \beta^h w_{i,t} + \sum_{k=1}^K \gamma_k^h w_{i,t-k} + \sum_{k=0}^K \delta_k^h y_{i,t-k} + \gamma^h X_{i,s,t} + u_{i,t}^h$$

where i , s , r , and t denote MSA, state, region, and month respectively. $h = 1, \dots, H$ denotes the estimation horizon. We control for lags up to $K = 3$. $y_{i,t}$ is year-on-year inflation of MSA i in month t measured by the logarithmic difference in the prices for services excluding housing, $\ln(\text{prices})_{i,t} - \ln(\text{prices})_{i,t-12}$. The dependent variable $y_{i,t}^h - y_{i,t}^0$ captures the difference in inflation over a horizon of h months from time t to time $t + h$. $\alpha_{r,t}^h$ is a region-time fixed effect. η_i^h is an MSA fixed effect. $w_{i,t}$ is our independent variable of interest, year-on-year wage growth in the service sector from Homebase measured by logarithmic difference, $\ln(\text{wage})_{i,t} - \ln(\text{wage})_{i,t-12}$. We instrument wage growth using the Bartik shock defined in equation (2) and constructed using tradable-sector industries following the definition in Hazell et al. (2022). We also include a vector of controls $X_{i,s,t}$ at the state-year level, including labor productivity (from BLS) to proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline and core excluding food and energy inflation, log household income per capita, log housing wealth defined as the product between Census homeownership rates and Freddie Mac House Price Index, and residuals from the regressions for previous horizons. The sample is a monthly panel of MSAs over January 2019 to December 2022. MSAs are mapped to states based on the principal state on which the MSA belongs to. States are categorized into four Census Bureau designated regions. Observations are weighted by the average size of the labor force in each MSA at 2019. We cluster the standard errors $u_{i,t}^h$ at the MSA level. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Source: BLS, Homebase, and authors' calculations.

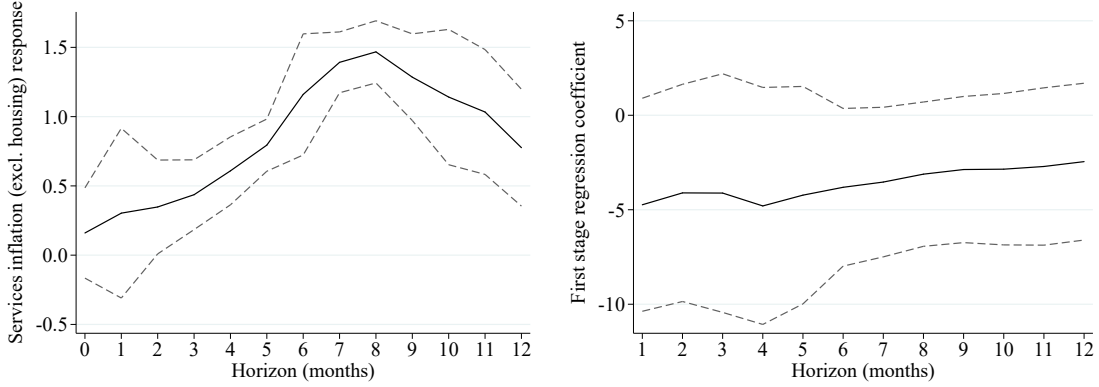


Figure 12: Services inflation (excluding housing) response, Bartik shock based on unemployment rate

Notes: This figure plots first and second stage estimates from the LP-IV model in equation (1):

$$y_{i,t}^h - y_{i,t}^0 = \alpha_t^h + \eta_i^h + \beta^h w_{i,t} + \sum_{k=1}^K \gamma_k^h w_{i,t-k} + \sum_{k=0}^K \delta_k^h y_{i,t-k} + \gamma^h X_{i,s,t} + u_{i,t}^h$$

where i , s , and t denote MSA, state, and month respectively. $h = 1, \dots, H$ denotes the estimation horizon. We control for lags up to $K = 3$. $y_{i,t}$ is year-on-year inflation of MSA i in month t measured by the logarithmic difference in the prices for services excluding housing, $\ln(prices)_{i,t} - \ln(prices)_{i,t-12}$. The dependent variable $y_{i,t}^h - y_{i,t}^0$ captures the difference in inflation over a horizon of h months from time t to time $t + h$. α_t^h is a time fixed effect. η_i^h is an MSA fixed effect. $w_{i,t}$ is our independent variable of interest, year-on-year wage growth in the service sector from Homebase measured by logarithmic difference, $\ln(wage)_{i,t} - \ln(wage)_{i,t-12}$. We instrument wage growth using a Bartik shock constructed using the log of unemployment rate in tradable-sector industries following the definition in Hazell et al. (2022). We also include a vector of controls $X_{i,s,t}$ at the state-year level, including labor productivity (from BLS) to proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline and core excluding food and energy inflation, and residuals from the regressions for previous horizons. The sample is a monthly panel of MSAs over January 2019 to December 2022. Observations are weighted by the average size of the labor force in each MSA at 2019. We cluster the standard errors $u_{i,t}^h$ at the MSA level. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Source: BLS, Homebase, and authors' calculations.

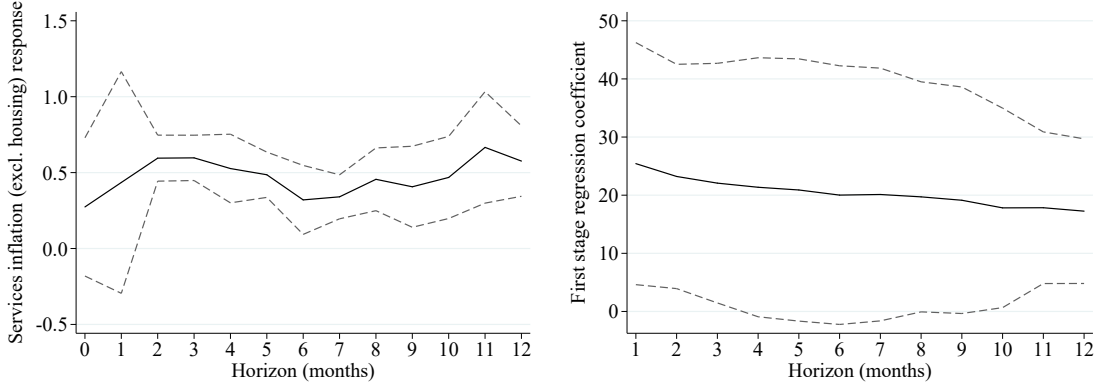


Figure 13: Services inflation (excluding housing) response, Bartik shock based on employment growth

Notes: This figure plots first and second stage estimates from the LP-IV model in equation (1):

$$y_{i,t}^h - y_{i,t}^0 = \alpha_t^h + \eta_i^h + \beta^h w_{i,t} + \sum_{k=1}^K \gamma_k^h w_{i,t-k} + \sum_{k=0}^K \delta_k^h y_{i,t-k} + \gamma^h X_{i,s,t} + u_{i,t}^h$$

where i , s , and t denote MSA, state, and month respectively. $h = 1, \dots, H$ denotes the estimation horizon. We control for lags up to $K = 3$. $y_{i,t}$ is year-on-year inflation of MSA i in month t measured by the logarithmic difference in the prices for services excluding housing, $\ln(\text{prices})_{i,t} - \ln(\text{prices})_{i,t-12}$. The dependent variable $y_{i,t}^h - y_{i,t}^0$ captures the difference in inflation over a horizon of h months from time t to time $t+h$. α_t^h is a time fixed effect. η_i^h is an MSA fixed effect. $w_{i,t}$ is our independent variable of interest, year-on-year wage growth in the service sector from Homebase measured by logarithmic difference, $\ln(\text{wage})_{i,t} - \ln(\text{wage})_{i,t-12}$. We instrument wage growth using a Bartik shock constructed using log employment growth in tradable-sector industries following the definition in Hazell et al. (2022). We also include a vector of controls $X_{i,s,t}$ at the state-year level, including labor productivity (from BLS) to proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline and core excluding food and energy inflation, and residuals from the regressions for previous horizons. The sample is a monthly panel of MSAs over January 2019 to December 2022. Observations are weighted by the average size of the labor force in each MSA at 2019. We cluster the standard errors $u_{i,t}^h$ at the MSA level. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Source: BLS, Homebase, and authors' calculations.

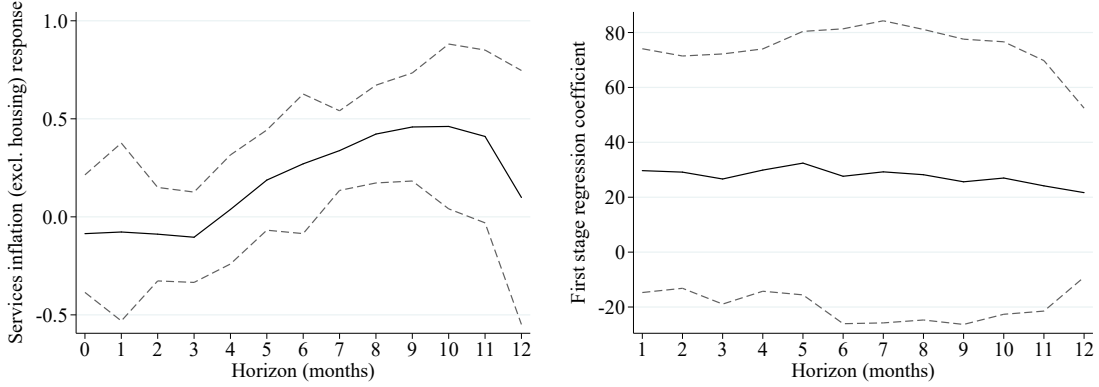


Figure 14: Services inflation (excluding housing) response, Bartik shock based on wage growth

Notes: This figure plots first and second stage estimates from the LP-IV model in equation (1):

$$y_{i,t}^h - y_{i,t}^0 = \alpha_t^h + \eta_i^h + \beta^h w_{i,t} + \sum_{k=1}^K \gamma_k^h w_{i,t-k} + \sum_{k=0}^K \delta_k^h y_{i,t-k} + \gamma^h X_{i,s,t} + u_{i,t}^h$$

where i , s , and t denote MSA, state, and month respectively. $h = 1, \dots, H$ denotes the estimation horizon. We control for lags up to $K = 3$. $y_{i,t}$ is year-on-year inflation of MSA i in month t measured by the logarithmic difference in the prices for services excluding housing, $\ln(\text{prices})_{i,t} - \ln(\text{prices})_{i,t-12}$. The dependent variable $y_{i,t}^h - y_{i,t}^0$ captures the difference in inflation over a horizon of h months from time t to time $t + h$. α_t^h is a time fixed effect. η_i^h is an MSA fixed effect. $w_{i,t}$ is our independent variable of interest, year-on-year wage growth in the service sector from Homebase measured by logarithmic difference, $\ln(\text{wage})_{i,t} - \ln(\text{wage})_{i,t-12}$. We instrument local labor market wage growth using a Bartik shock constructed using log wage growth in tradable-sector industries following the definition in Hazell et al. (2022). We also include a vector of controls $X_{i,s,t}$ at the state-year level, including labor productivity (from BLS) to proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline and core excluding food and energy inflation, and residuals from the regressions for previous horizons. The sample is a monthly panel of MSAs over January 2019 to December 2022. Observations are weighted by the average size of the labor force in each MSA at 2019. We cluster the standard errors $u_{i,t}^h$ at the MSA level. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Source: BLS, Homebase, and authors' calculations.

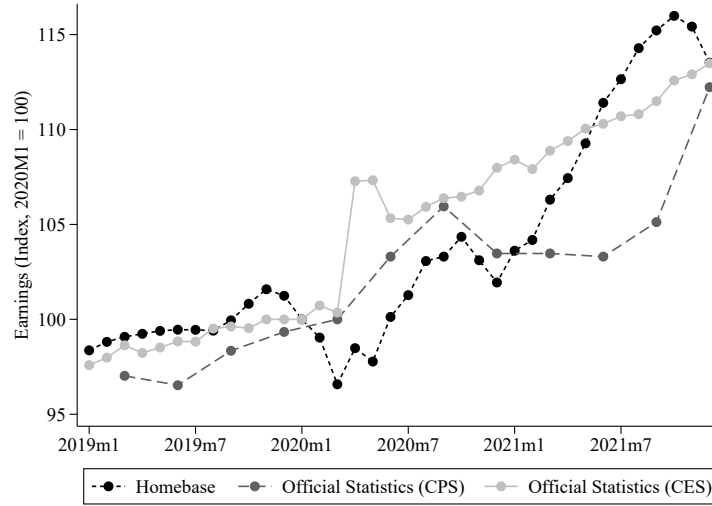


Figure 15: Average wage trend: a comparison

Notes: This figure plots national average earnings in the Homebase sample against official statistics. We consider two sources of official statistics, both from the BLS. The Current Population Survey (CPS) line plots the median weekly nominal earnings of wage and salary workers in service occupations of 16 years and over (LEU0254543400Q). The Current Employment Statistics (CES) line plots the average weekly earnings of all employees from the private service-providing sector (CES0800000011). The Homebase sample excludes workers that were on temporary layoffs, defined as workers that were initially employed (reported positive earnings), were laid off during the middle of the sample (reported zero earnings), and later returned to work again before the end of the sample. The sample includes all workers that were in the database for at least one month between 2019 and 2021. Each series is expressed as indices relative to its value in January 2020 (and Q1 2020 for CPS). The correlation between Homebase and CPS series is 0.73; the correlation between Homebase and CES series is 0.87. Source: BLS, Homebase, and authors' calculations.

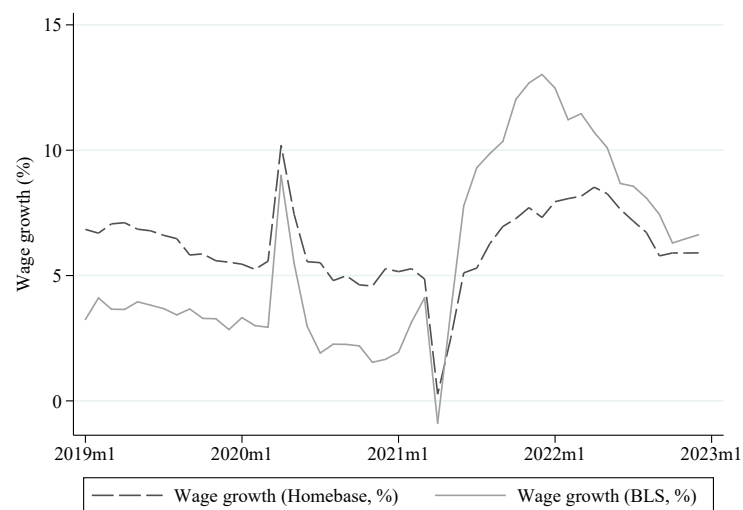


Figure 16: Wage growth: a comparison

Notes: This figure plots year-on-year average hourly wage growth from Homebase and the average hourly earnings for the leisure and hospitality industry (CES7000000003) from the Current Employment Statistics (CES). The Homebase and CES series have a correlation of 0.68. Source: BLS, Homebase, and authors' calculations.

References

- Afrouzi, Hassan, Andrés Blanco, Andrés Drenik, and Erik Hurst**, “A Theory of How Workers Keep Up With Inflation,” Working Paper 33233, National Bureau of Economic Research December 2024.
- Amiti, Mary, Sebastian Heise, Fatih Karahan, and Ayşegül Şahin**, “Inflation Strikes Back: The Role of Import Competition and the Labor Market,” *NBER Macroeconomics Annual*, None 2024, 38 (1), 71–131.
- Bagga, Sadhika, Lukas Friedrich Mann, Ayşegül Şahin, and Giovanni L Violante**, “Job Amenity Shocks and Labor Reallocation,” Working Paper 33787, National Bureau of Economic Research May 2025.
- Ball, L., D. Leigh, and P. Mishra**, “Understanding US Inflation during the COVID Era,” NBER Working Papers 30613, NBER 2022.
- Barnichon, Régis**, “Building a Composite Help-Wanted Index,” *Economics Letters*, 2010, 109 (3), 175–178.
- Barnichon, Régis and Adam Hale Shapiro**, “Phillips meets Beveridge,” *Journal of Monetary Economics*, 2024, 148, 103660. Inflation in the COVID Era and Beyond.
- Bartik, Timothy J.**, *Who Benefits from State and Local Economic Development Policies?* number wbsle. In ‘Books from Upjohn Press.’, W.E. Upjohn Institute for Employment Research, November 1991.
- Benigno, Pierpaolo and Gauti B Eggertsson**, “It’s Baaack: The Surge in Inflation in the 2020s and the Return of the Non-Linear Phillips Curve,” NBER Working Papers 31197, NBER April 2023.
- Beraja, Martin, Erik Hurst, and Juan Ospina**, “The Aggregate Implications of Regional Business Cycles,” *Econometrica*, 2019, 87 (6), 1789–1833.
- Bernanke, Ben S. and Olivier J Blanchard**, “An Analysis of Pandemic-Era Inflation in 11 Economies,” Working Paper Series WP24-11, Peterson Institute for International Economics May 2024.
- and —, “What Caused the US Pandemic-Era Inflation?,” *American Economic Journal: Macroeconomics*, 2024.
- Blanchard, Olivier, Alex Domash, and Lawrence Summers**, “Bad News for the Fed from the Beveridge Space,” *Peterson Institute for International Economics Policy Brief*, 2022, 22 (7).
- Borusyak, Kirill, Peter Hull, and Xavier Jaravel**, “Quasi-Experimental Shift-Share Research Designs: Sampling-based vs. Design-based Uncertainty in Regression Analysis,” *The Review of Economic Studies*, 2022, 89 (1), 181–213.
- Chen, Sophia and Do Lee**, “Divergence in Post-Pandemic Earnings Growth: Evidence from Micro Data,” *IMF Working Paper*, 2024.
- Crump, Richard K, Stefano Eusepi, Marc Giannoni, and Ayşegül Şahin**, “The Unemployment-Inflation Trade-off Revisited: The Phillips Curve in COVID Times,” NBER Working Papers 29785, NBER February 2022.
- Dao, Mai Chi, Pierre-Olivier Gourinchas, Prachi Mishra, and Daniel Leigh**, “Understanding the International Rise and Fall of Inflation Since 2020,” *Journal of Monetary Economics*, 2024.
- Domash, Alex and Lawrence H. Summers**, “A labor market view on the risks of a U.S. hard landing,” *Journal of Policy Modeling*, 2022, 44 (4), 758–767. WHERE IS THE US AND WORLD ECONOMY HEADED?
- Dvorkin, M. A. and M. Isaacson**, “Tracking Wage Inflation in Real Time,” *Federal Reserve Bank of St Louis, On the Economy Blog*, 2022.

- Furman, Jason and Powell Wilson III**, “What is the Best Measure of Labor Market Tightness?,” Peterson Institute for International Economics blog November 22, Peterson Institute for International Economics 2021.
- Guerreiro, Joao, Jonathon Hazell, Chen Lian, and Christina Patterson**, “Why Do Workers Dislike Inflation? Wage Erosion and Conflict Costs,” Working Paper 32956, National Bureau of Economic Research September 2024.
- Hazell, Jonathon, Juan Herreño, Emi Nakamura, and Jón Steinsson**, “The Slope of the Phillips Curve: Evidence from U.S. States,” *The Quarterly Journal of Economics*, 2022, *123* (3), 1299–1344.
- Hooper, Peter, Frederic S. Mishkin, and Amir Sufi**, “Prospects for Inflation in a High Pressure Economy: Is the Phillips Curve Dead or is it Just Hibernating?,” *Research in Economics*, 2020, *74* (1), 26–62.
- Howard, Greg, Jack Liebersohn, and Adam Ozimek**, “The Short- and Long-run Effects of Remote Work on U.S. Housing Markets,” *Journal of Financial Economics*, 2023, *150* (1), 166–184.
- Johnson, Noah N.**, “Tradable and Nontradable Inflation Indexes: Replicating New Zealand’s Tradable Indexes with BLS CPI Data,” Monthly Labor Review, U.S. Bureau of Labor Statistics 2017.
- Jordà, Òscar**, “Estimation and Inference of Impulse Responses by Local Projections,” *American economic review*, 2005, *95* (1), 161–182.
- LaBelle, Jesse and Ana Maria Santacreu**, “Global Supply Chain Disruptions and Inflation During the COVID-19 Pandemic,” *Review*, 2022, *104* (2), 78–91.
- McLeay, Michael and Silvana Tenreyro**, “Optimal Inflation and the Identification of the Phillips Curve,” *NBER Macroeconomics Annual*, 2019, *34*.
- Pilossof, Laura and Jane M Ryngaert**, “Job Search, Wages, and Inflation,” Working Paper 33042, National Bureau of Economic Research October 2024.
- Soh, Jiaming, Myrto Oikonomou, Carlo Pizzinelli, Ippei Shibata, and Marina Mendes Tavares**, “Did the COVID-19 Recession Increase the Demand for Digital Occupations in the United States? Evidence from Employment and Vacancies Data,” IMF Working Papers 2022/195, International Monetary Fund 2022.
- Stewart, Jay**, “Why was Labor Productivity Growth So High during the COVID-19 Pandemic? The Role of Labor Composition,” BLS Working Papers 545, U.S. Bureau of Labor Statistics 2022.
- Stock, James H. and Mark W. Watson**, “Slack and Cyclically Sensitive Inflation,” *Journal of Money, Credit and Banking*, December 2020, *52* (S2), 393–428.
- Teulings, Coen N. and Nikolay Zubanov**, “Is Economic Recovery a Myth? Robust Estimation of Impulse Responses,” *Journal of Applied Econometrics*, 2014, *29* (3), 497–514.
- Òscar Jordà, Moritz Schularick, and Alan M. Taylor**, “Betting the House,” *Journal of International Economics*, 2015, *96* (Supplement 1), S2–S18.
- , —, and —, “The Effects of Quasi-Random Monetary Experiments,” *Journal of Monetary Economics*, 2020, *112*, 381–419.

A Appendix

A.1 Cross-sectional evidence from official statistics

We estimate our LP-IV model using wage data from the QCEW at the MSA quarterly level:

$$y_{i,t}^h - y_{i,t}^0 = \alpha_t^h + \eta_i^h + \beta^h w_{i,t} + \sum_{k=1}^K \gamma_k^h w_{i,t-k} + \sum_{k=0}^K \delta_k^h y_{i,t-k} + \gamma^h X_{i,s,t} + u_{i,t}^h \quad (4)$$

where i , s , and t denote MSA, state, and quarter, respectively. $h = 1, \dots, H$ denotes the estimation horizon. We control for lags up to $K = 1$. $y_{i,t}$ is year-on-year inflation of MSA i in quarter t measured by the logarithmic difference in the prices for services excluding housing, $\ln(prices)_{i,t} - \ln(prices)_{i,t-4}$. The dependent variable $y_{i,t}^h - y_{i,t}^0$ captures the difference in inflation over a horizon of h quarters starting at time t . α_t^h is a time fixed effect. η_i^h is an MSA fixed effect. $w_{i,t}$ is our independent variable of interest, year-on-year wage growth in the service sector from the QCEW measured by logarithmic difference, $\ln(wage)_{i,t} - \ln(wage)_{i,t-4}$. We instrument wage growth using the Bartik shock for the log vacancy-to-unemployment ratio, which is defined in equation (2), constructed using only tradable-sector industries following the definition in Hazell et al. (2022), where the monthly shock is converted to quarterly frequencies by taking the average within each calendar quarter. We also include a vector of controls $X_{i,s,t}$ at the state-year level, including labor productivity (from BLS) to proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline and core excluding food and energy inflation, and residuals from the regressions for previous horizons. Observations are weighted by the average size of the labor force in each MSA at 2019. We cluster the standard errors $u_{i,t}^h$ at the MSA level. The sample period is from Q1 2019 to Q4 2022.¹⁵ Figure 17 plots the estimated coefficients and Figure 18 decomposes estimation results at the 1-quarter horizon.

A.2 Time-series evidence from official statistics

We use official statistics from BLS on vacancy, unemployment, and prices at the national level to estimate a dynamic Phillips curve using the following LP specification

$$y_t^h = \alpha^h + \beta^h \ln \theta_t + \sum_{k=1}^K \gamma_{\theta,k}^h \ln \theta_{t-k} + \sum_{k=1}^K \gamma_{y,k}^h y_{t-k} + \gamma_x^h X_t + u_t^h$$

for monthly time t and horizon $h = 0, 1, \dots, 36$ months. y_t is year-on-year services (excluding housing) inflation in month t , computed as logarithmic changes. The dependent variable y_t^h denotes inflation h months ahead of time t . $\ln \theta_t$ is our independent variable of interest, the log vacancy-to-unemployment ratio from Barnichon (2010) and updated from JOLTS. X_t is a vector of control variables including labor productivity (from BLS), headline inflation shocks, and inflation expectations. Headline inflation shock is defined as the difference between year-on-year headline and core excluding food and energy inflation. Inflation expectations are measured using the 2-year inflation expectations from the Federal Reserve Bank of Cleveland, combined with 12-month inflation expectations from the Livingston survey before 1982. We include $K = 3$ lags of the dependent variable and each independent variable. The sample period is January 1986 to December 2023. Figure 19 plots the estimated coefficients.

We explore the time-variation in this parameter by keeping the horizon fixed at $h = 12$ months (the peak of the estimation above) and running rolling regressions with a 3-year window. Figure 20 plots the estimated coefficients. We find substantial time-variation in the coefficient. The estimate for the 2021-2023 period is around 4.0. This is quantitatively very similar to the estimates in Benigno and Eggertsson (2023) during periods with a high vacancy-to-unemployment ratio.

To illustrate the important role of labor market tightness for services (excluding housing) inflation, Figure 21 plots the fitted values (the “Fitted” line) from the rolling regression above as well as the fitted value (the “Fitted w/o $\ln(V/U)$ ” line) from the same regression without a tight labor market by imposing that V/U equal 1. It is evident that the “Fitted” line offers a much better fit of actual inflation than the “Fitted w/o $\ln(V/U)$ ” line, indicating an important role of labor market tightness in explaining the recent

¹⁵Wage data start from Q1 1990 but the Bartik shock data only starts in Q1 2019.

inflation surge, especially after mid-2021. Without a tight labor market, the services (excluding housing) inflation would have peaked in early 2021 instead of early 2022 and would have fallen under 2 percent, instead of the actual rate of 4 percent, in early 2023.

A.3 Constructing nationally representative vacancies

Data on vacancies from Indeed may not necessarily be nationally representative because some job openings may not be posted online. To address this limitation, we rescale the Indeed vacancies data by applying the following procedure. Nationally representative data are available for unemployment from BLS's CPS (2- and 3-digit NAICS industry level) and vacancies from BLS's JOLTS (2-digit NAICS industry level). To obtain nationally representative vacancies at the 3-digit NAICS level, we first calculate the log vacancy-to-unemployment ratio $\ln(V_{k^3,t}/U_{k^3,t})$ for each k^3 at the 3-digit NAICS level using data on vacancies from Indeed and data on unemployment from the CPS. We regress the 3-digit NAICS level log vacancy-to-unemployment ratio $\ln(V_{k^3,t}/U_{k^3,t})$ onto the corresponding 2-digit NAICS level log vacancy-to-unemployment ratio $\ln(V_{k^2,t}/U_{k^2,t})$, where the 2-digit NAICS level data use vacancies from Indeed and unemployment from the CPS. We obtain the fitted values of the regression to obtain a nationally representative 3-digit NAICS level log vacancy-to-unemployment ratio $\ln(\widehat{V_{k^3,t}/U_{k^3,t}})$. Finally, we add back the 3-digit log unemployment from the denominator of the ratio and then exponentiate the expression to convert the fitted values of log vacancy-to-unemployment ratio into nationally representative 3-digit vacancies: $\exp(\ln(\widehat{V_{k^3,t}/U_{k^3,t}}) + \ln(U_{k^3,t}))$.

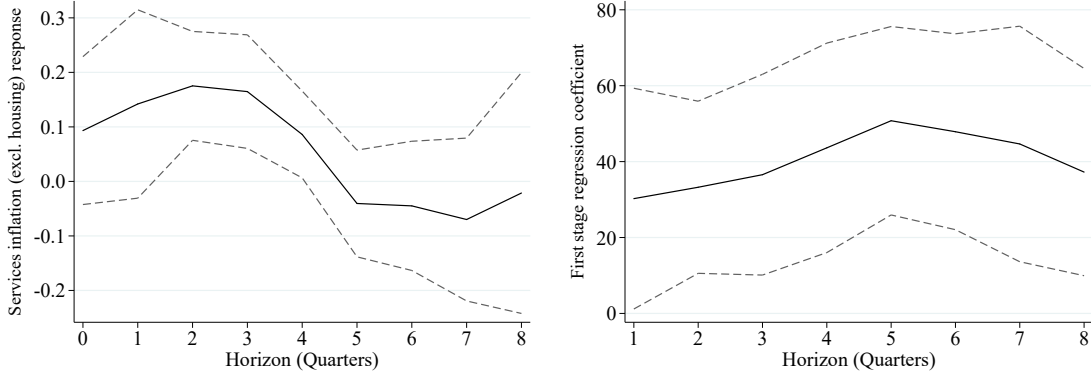


Figure 17: Services inflation (excluding housing) response, MSA level using QCEW wages

Notes: This figure plots first and second stage estimates from the following LP-IV model:

$$y_{i,t}^h - y_{i,t}^0 = \alpha_t^h + \eta_i^h + \beta^h w_{i,t} + \sum_{k=1}^K \gamma_k^h w_{i,t-k} + \sum_{k=0}^K \delta_k^h y_{i,t-k} + \gamma^h X_{i,s,t} + u_{i,t}^h$$

where i , s , and t denote MSA, state, and quarter, respectively. $h = 1, \dots, H$ denotes the estimation horizon. We control for lags up to $K = 1$. $y_{i,t}$ is year-on-year inflation of MSA i in quarter t measured by the logarithmic difference in the prices for services excluding housing, $\ln(prices)_{i,t} - \ln(prices)_{i,t-4}$. The dependent variable $y_{i,t}^h - y_{i,t}^0$ captures the difference in inflation over a horizon of h quarters starting at time t . α_t^h is a time fixed effect. η_i^h is an MSA fixed effect. $w_{i,t}$ is our independent variable of interest, year-on-year wage growth in the service sector from the QCEW measured by logarithmic difference, $\ln(wage)_{i,t} - \ln(wage)_{i,t-4}$. We instrument wage growth using the Bartik shock for the log vacancy-to-unemployment ratio, which is defined in equation (2), constructed using only tradable-sector industries following the definition in Hazell et al. (2022). We also include a vector of controls $X_{i,s,t}$ at the state-year level, including labor productivity (from BLS) to proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline and core excluding food and energy inflation, and residuals from the regressions for previous horizons. The sample is a quarterly panel of MSAs over Q1 2019 to Q4 2022. Observations are weighted by the average size of the labor force in each MSA at 2019. We cluster the standard errors $u_{i,t}^h$ at the MSA level. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Source: BLS, QCEW, and authors' calculations.

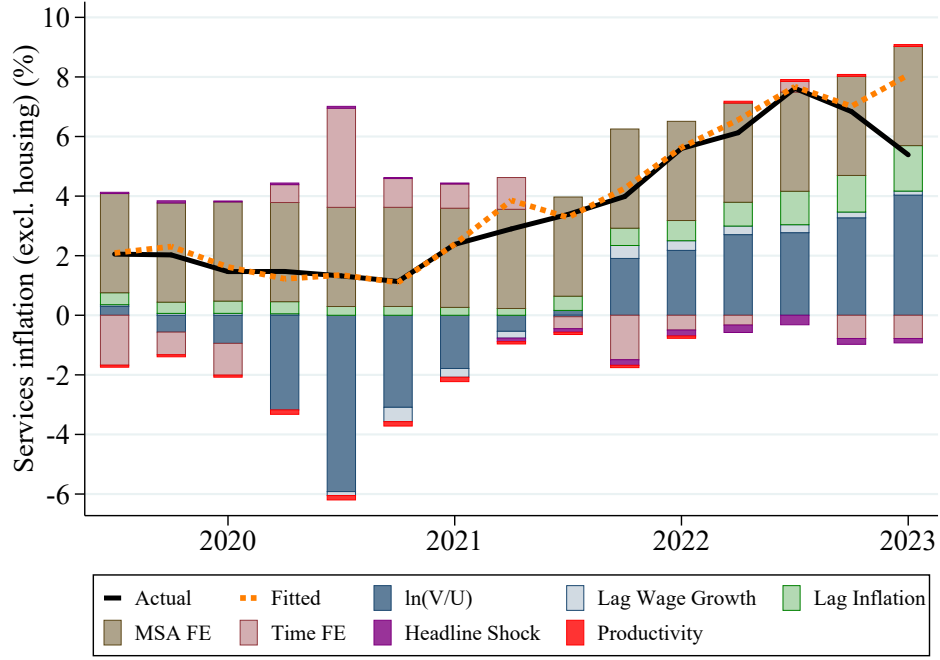


Figure 18: Predictions for services inflation (excluding housing), MSA level using QCEW wages

Notes: This figure plots fitted values from the LP-IV model estimated at the MSA level (Figure 17):

$$y_{i,t}^h = \alpha_{2,t}^h + \eta_{2,i}^h + \beta_2^h \hat{w}_{i,t} + \sum_{k=1}^K \gamma_{2,k}^h w_{i,t-k} + \sum_{k=0}^K \delta_{2,k}^h y_{i,t-k} + \gamma_2^h X_{i,s,t} + u_{2,i,t}^h$$

$$w_{i,t} = \alpha_{1,t}^h + \eta_{1,i}^h + \beta_1^h Shock_{i,t} + \sum_{k=1}^K \gamma_{1,k}^h w_{i,t-k} + \sum_{k=0}^K \delta_{1,k}^h y_{i,t-k} + \gamma_1^h X_{i,s,t} + u_{1,i,t}^h$$

where i , s , and t denote MSA, state, and quarter, respectively. $h = 1$ denotes the estimation horizon. The solid line plots actual inflation, and the dashed line plots fitted values from the LP-IV model. All values are unweighted averages across MSAs in the sample. The bars show the contribution of each independent variable in the LP-IV model, where the contribution captures the combined effect of each variable from the first and second stages. “ln(V/U)” plots the fitted value from the Bartik shock $Shock_{i,t}$ for the log vacancy-to-unemployment ratio defined in equation (2), constructed using only tradable-sector industries following the definition in Hazell et al. (2022). “Lag Wage Growth” contains fitted values from $w_{i,t-k}$, “Lag Inflation” from $y_{i,t-k}$, “MSA FE” from $\alpha_{1,t}^h$ and $\alpha_{2,t}^h$, “Time FE” from $\eta_{1,t}^h$ and $\eta_{2,t}^h$, and “Headline Shock” and “Productivity” from the controls in $X_{i,s,t}$. Source: BLS, Homebase, and authors’ calculations.

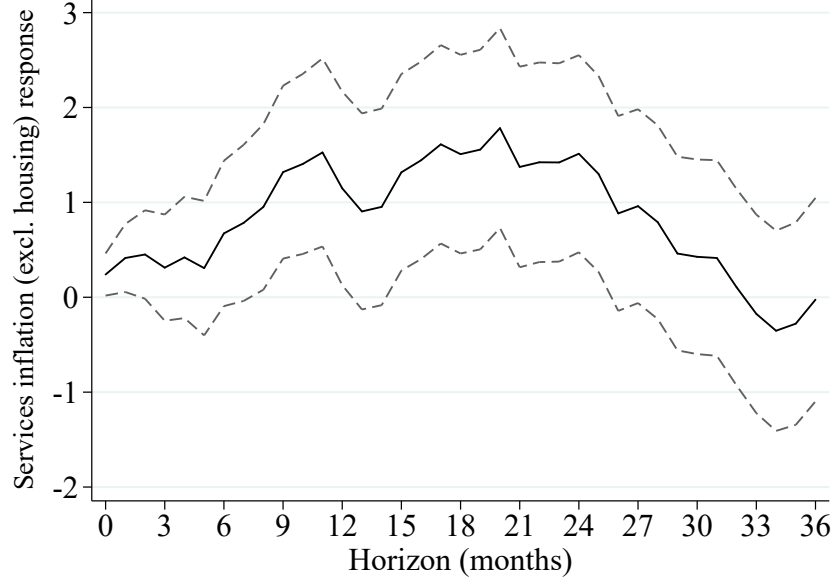


Figure 19: Services inflation (excluding housing) response, national level official statistics

Notes: This figure plots estimates of β^h from the following local projection (LP) model:

$$y_t^h = \alpha^h + \beta^h \ln \theta_t + \sum_{k=1}^K \gamma_{\theta,k}^h \ln \theta_{t-k} + \sum_{k=1}^K \gamma_{y,k}^h y_{t-k} + \gamma_x^h X_t + u_t^h$$

for monthly time t and horizon $h = 0, 1, \dots, 36$ months. y_t is year-on-year CPI services inflation (excluding housing) in month t , computed as logarithmic changes. The dependent variable y_t^h denotes inflation h months ahead of time t . $\ln \theta_t$ is our independent variable of interest, the log vacancy-to-unemployment ratio from Barnichon (2010) and updated from JOLTS. X_t is a vector of control variables including labor productivity (from BLS), headline inflation shocks, and inflation expectations. Headline inflation shock is defined as the difference between year-on-year headline and core excluding food and energy inflation. Inflation expectations are measured using the 2-year inflation expectations from the Federal Reserve Bank of Cleveland, combined with 12-month inflation expectations from the Livingston survey before 1982. We include $K = 3$ lags of the dependent variable and each independent variable. The sample period is January 1986 to December 2023. Standard errors are Newey-West. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Source: BLS, Homebase, and authors' calculations.

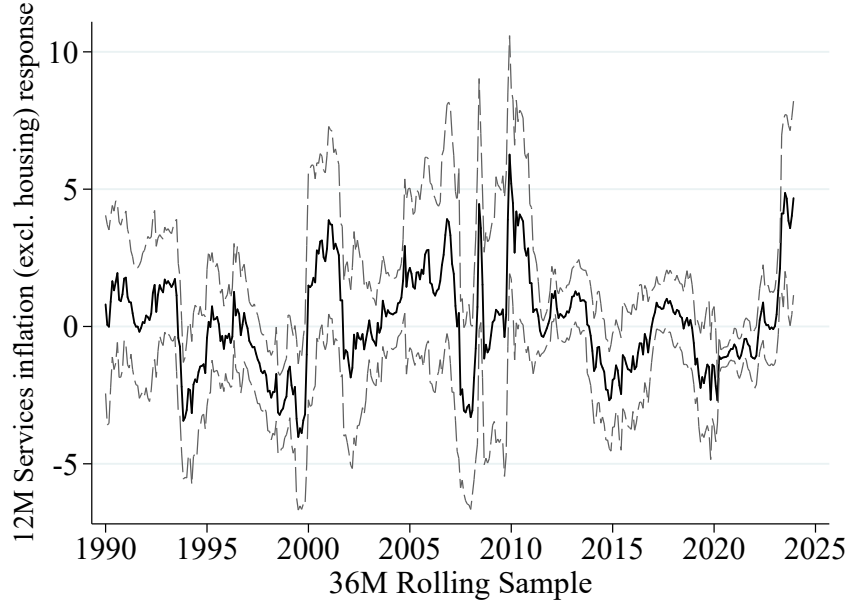


Figure 20: 12-month services inflation (excluding housing) response, 3-year rolling sample, national level official statistics

Notes: This figure plots rolling estimates from the following local projection (LP) model:

$$y_t^h = \alpha^h + \beta^h \ln \theta_t + \sum_{k=1}^K \gamma_{\theta,k}^h \ln \theta_{t-k} + \sum_{k=1}^K \gamma_{y,k}^h y_{t-k} + \gamma_x^h X_t + u_t^h$$

for monthly time t and horizon $h = 3$ months. The regression is estimated rolling samples with a backward-looking 3-year window, and the x axis denotes the last month in each rolling sample. y_t is year-on-year CPI services inflation (excluding housing) in month t , computed as logarithmic changes. The dependent variable y_t^h denotes inflation h months ahead of time t . $\ln \theta_t$ is our independent variable of interest, the log vacancy-to-unemployment ratio from Barnichon (2010) and updated from JOLTS. X_t is a vector of control variables including labor productivity (from BLS), headline inflation shocks, and inflation expectations. Headline inflation shock is defined as the difference between year-on-year headline and core excluding food and energy inflation. Inflation expectations are measured using the 2-year inflation expectations from the Federal Reserve Bank of Cleveland, combined with 12-month inflation expectations from the Livingston survey before 1982. We include $K = 3$ lags of the dependent variable and each independent variable. Standard errors are Newey-West. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Source: BLS, Homebase, and authors' calculations.

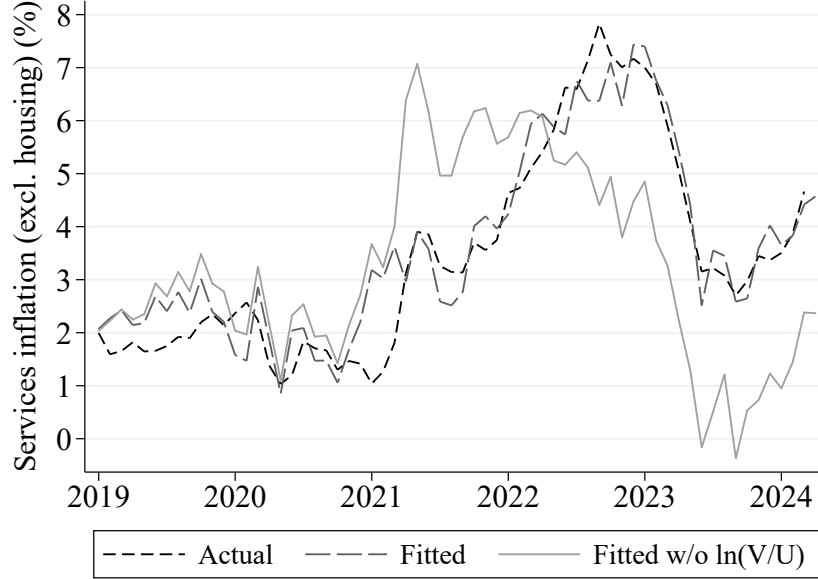


Figure 21: Predictions for services inflation (excluding housing), national level official statistics

Notes: This figure plots fitted values from the local projection (LP) model estimated over 3-year rolling samples (Figure 19):

$$y_t^h = \alpha^h + \beta^h \ln \theta_t + \sum_{k=1}^K \gamma_{\theta,k}^h \ln \theta_{t-k} + \sum_{k=1}^K \gamma_{y,k}^h y_{t-k} + \gamma_x^h X_t + u_t^h$$

for monthly time t and horizon $h = 3$ months. The regression is estimated rolling samples with a backward-looking 3-year window. “Fitted” line plots predictions from the full model. “Fitted w/o $\ln(V/U)$ ” line plots predictions from the same regression without a tight labor market by imposing V/U equals 1. y_t is year-on-year CPI services inflation (excluding housing) in month t , computed as logarithmic changes. The dependent variable y_t^h denotes inflation h months ahead of time t . $\ln \theta_t$ is our independent variable of interest, the log vacancy-to-unemployment ratio from Barnichon (2010) and updated from JOLTS. X_t is a vector of control variables including labor productivity (from BLS), headline inflation shocks, and inflation expectations. Headline inflation shock is defined as the difference between year-on-year headline and core excluding food and energy inflation. Inflation expectations are measured using the 2-year inflation expectations from the Federal Reserve Bank of Cleveland, combined with 12-month inflation expectations from the Livingston survey before 1982. We include $K = 3$ lags of the dependent variable and each independent variable. Source: BLS, Homebase, and authors’ calculations.

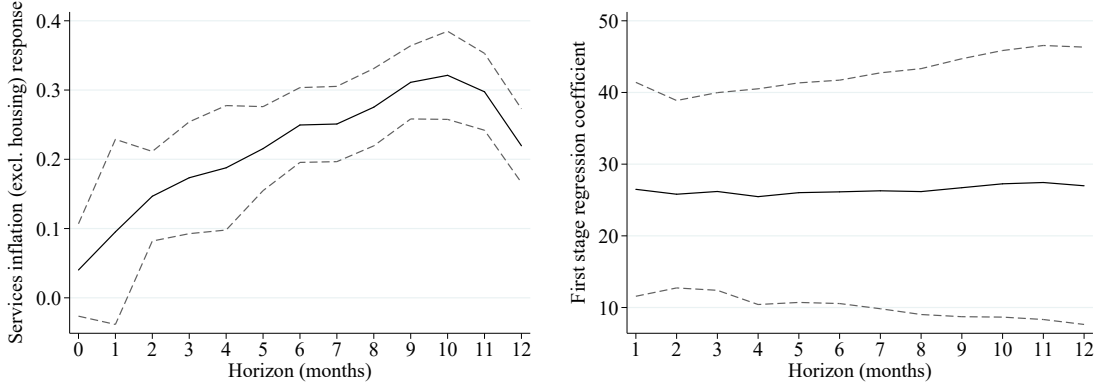


Figure 22: Services inflation (excluding housing) response, Bartik instrument using all sectors

Notes: This figure plots first and second stage estimates from the LP-IV model in equation (1):

$$y_{i,t}^h - y_{i,t}^0 = \alpha_t^h + \eta_i^h + \beta^h w_{i,t} + \sum_{k=1}^K \gamma_k^h w_{i,t-k} + \sum_{k=0}^K \delta_k^h y_{i,t-k} + \gamma^h X_{i,s,t} + u_{i,t}^h$$

where i , s , and t denote MSA, state, and month respectively. $h = 1, \dots, H$ denotes the estimation horizon. We control for lags up to $K = 3$. $y_{i,t}$ is year-on-year inflation of MSA i in month t measured by the logarithmic difference in the prices for services excluding housing, $\ln(prices)_{i,t} - \ln(prices)_{i,t-12}$. The dependent variable $y_{i,t}^h - y_{i,t}^0$ captures the difference in inflation over a horizon of h months from time t to time $t + h$. α_t^h is a time fixed effect. η_i^h is an MSA fixed effect. $w_{i,t}$ is our independent variable of interest, year-on-year wage growth in the service sector from Homebase measured by logarithmic difference, $\ln(wage)_{i,t} - \ln(wage)_{i,t-12}$. We instrument wage growth using the Bartik shock for the log vacancy-to-unemployment ratio, which is defined in equation (2), constructed using all 3-digit NAICS industries. We also include a vector of controls $X_{i,s,t}$ at the state-year level, including labor productivity (from BLS) to proxy for local labor demand, headline inflation shocks defined as the difference between year-on-year headline and core excluding food and energy inflation, and residuals from the regressions for previous horizons. The sample is a monthly panel of MSAs over January 2019 to December 2022. Observations are weighted by the average size of the labor force in each MSA at 2019. We cluster the standard errors $u_{i,t}^h$ at the MSA level. The solid line shows the point estimates and the dashed lines show the 90 percent confidence interval. Source: BLS, Homebase, and authors' calculations.